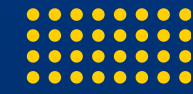


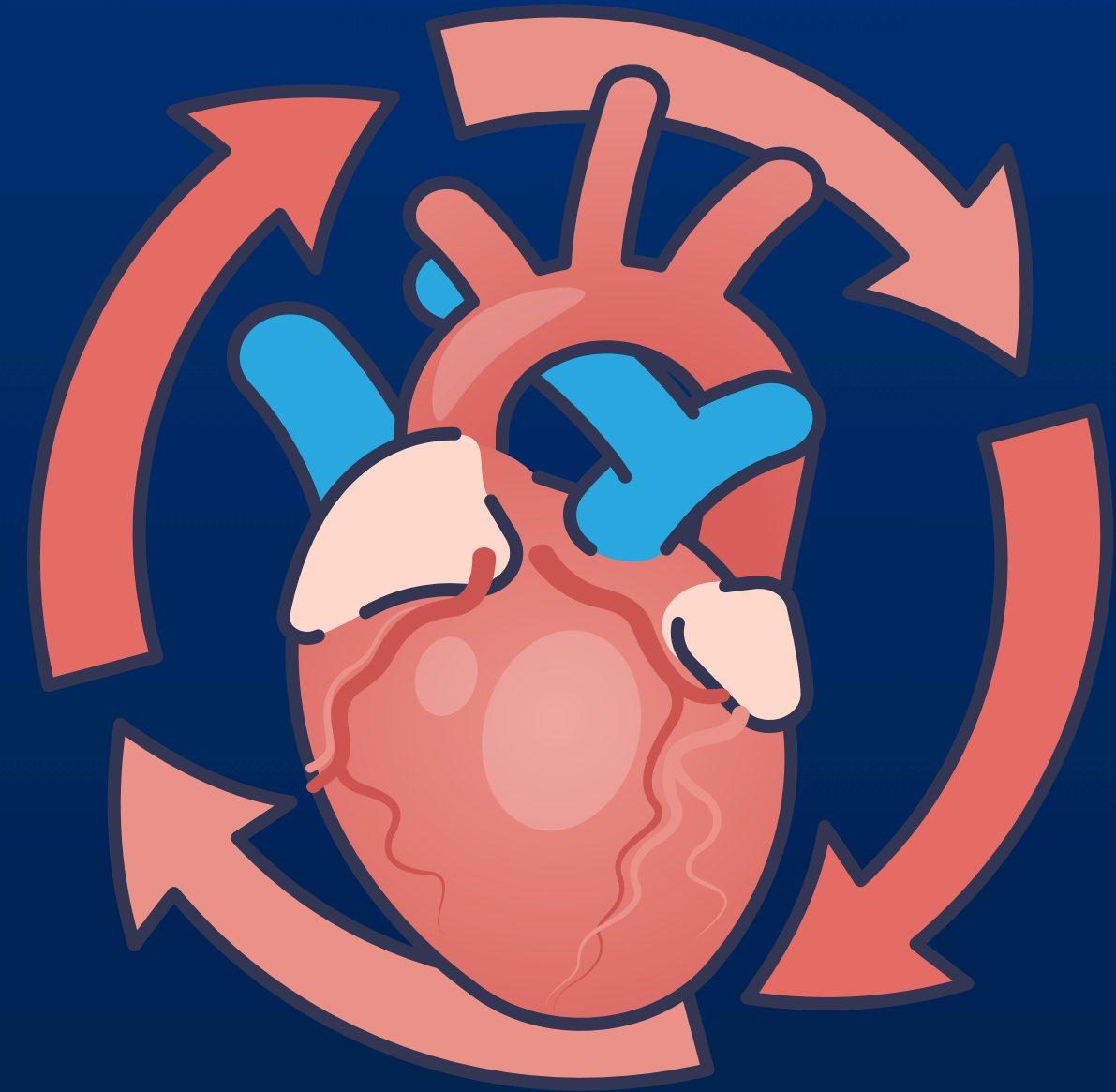
Cardiac cycle and cardiac output



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LEARNING OBJECTIVES

- 1. Sequence of electrical activity and mechanical activity in a cardiac cycle**
- 2. Pressures, volume and flow in the heart and great vessels in a cardiac cycle**
- 3. Heart sound and murmurs**
- 4. Definition and significance of cardiac output**
- 5. Regulation of cardiac output**
 - 5.1 heart rate and factors affecting heart rate
 - 5.2 stroke volume and factors affecting stroke volume
- 6. Definition and significance of the venous return**
- 7. Factors that influence venous return**
 - 7.1 postural changes
 - 7.2 skeletal muscle pump
 - 7.3 venacava compression
 - 7.4 respiration activity
- 8. Alterations in cardiac output and venous return due to the pathology of the cardiovascular system and extracellular fluid electrolyte abnormalities**
 - 8.1 Effects of hypercalcemia and hypocalcemia on cardiac cycle
 - 8.2 Effects of hyperkalemia and hypokalemia on cardiac cycle

1. Sequence of **electrical activity** and **mechanical activity** in a cardiac cycle

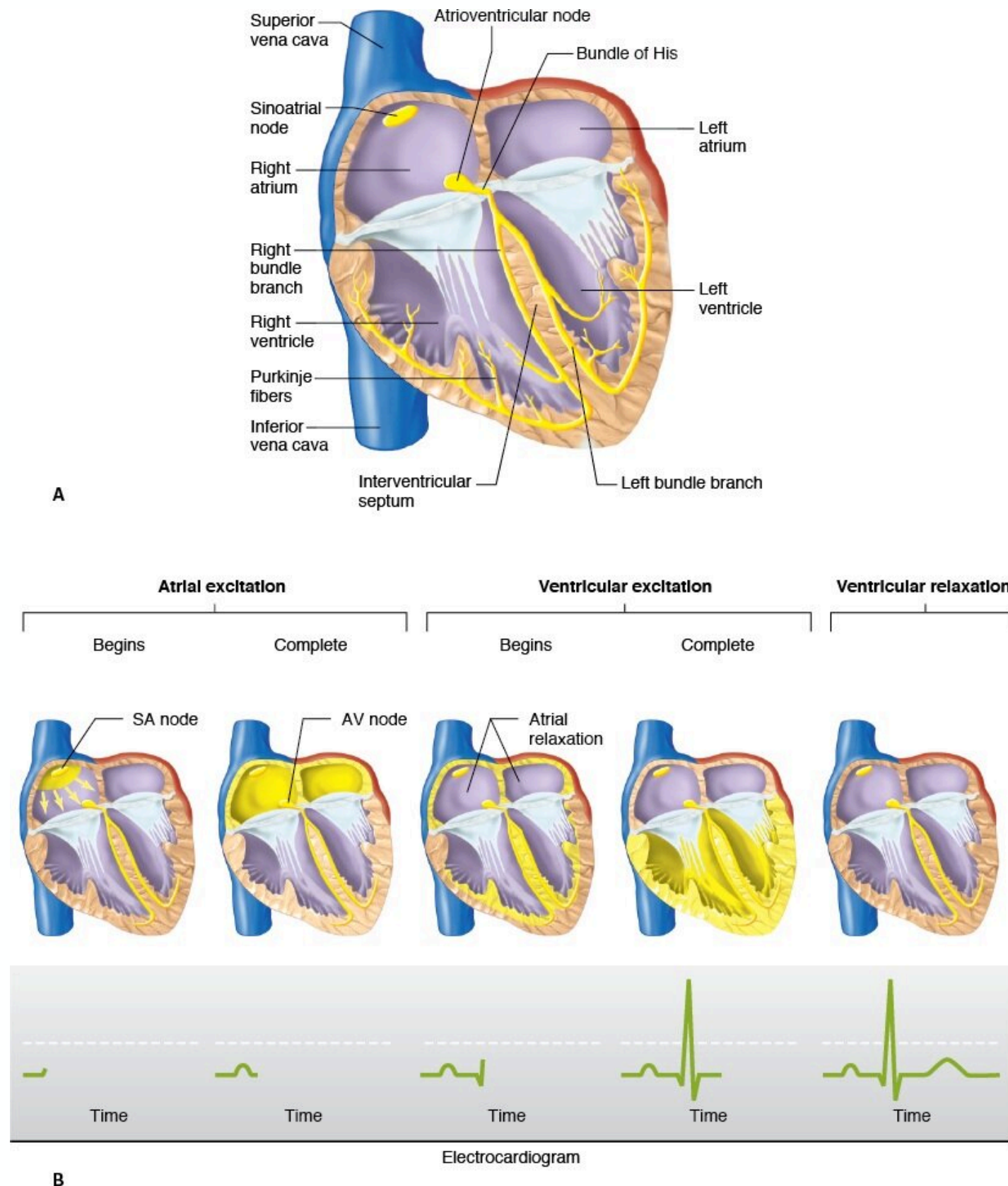
- The signal travels from SA node to the AV node, and from there to the right and left ventricles, which also contract together, just a split second after the atria contract.
- Each part of the cardiac cycle consists of several phases:.

Diastole includes:

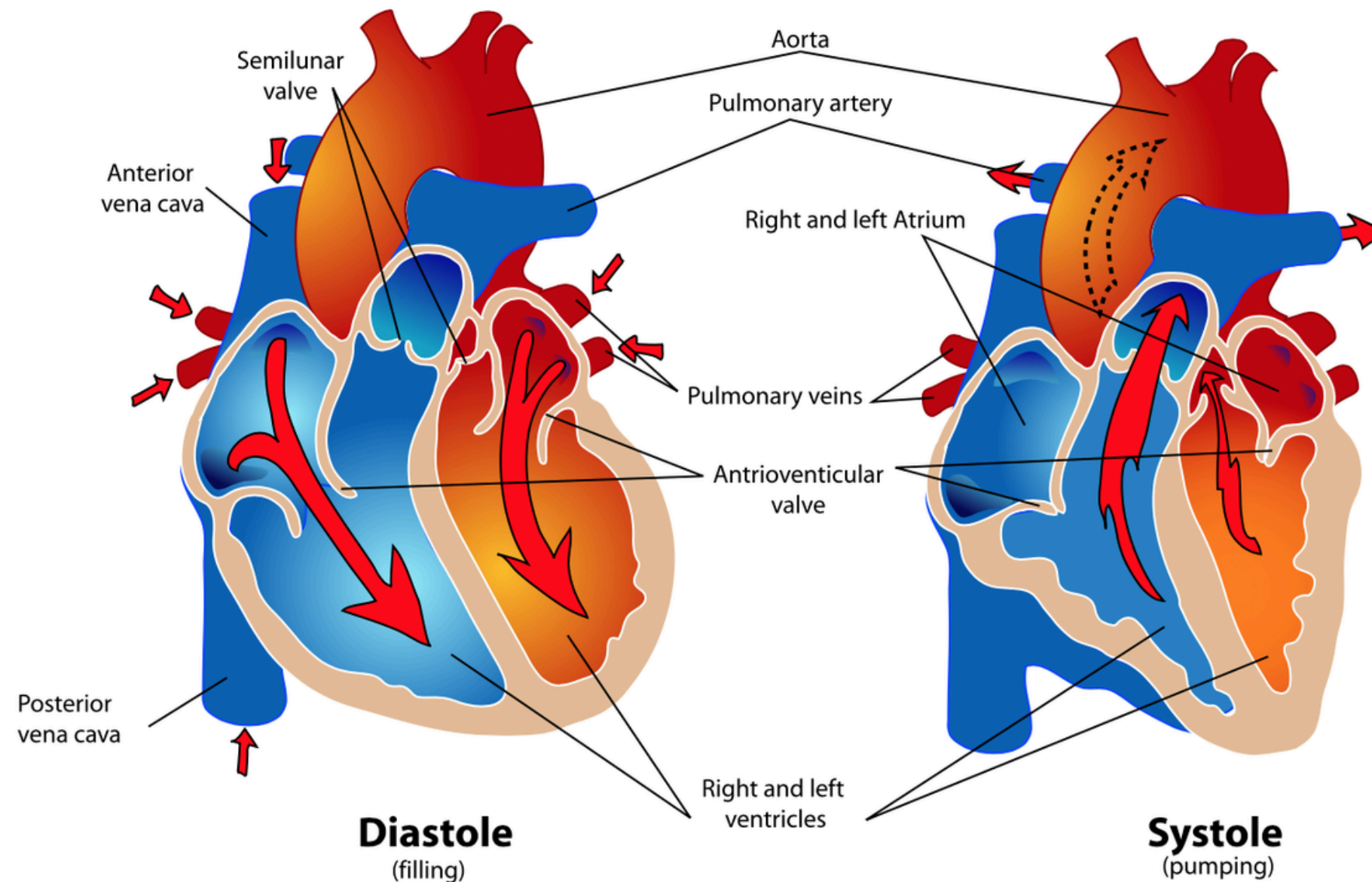
- Isovolumic relaxation
- Atrial contraction
- Rapid ventricular filling
- Slow ventricular filling (diastasis)

Systole includes:

- Isovolumic contraction
- Ejection



1. Sequence of electrical activity and mechanical activity in a cardiac cycle

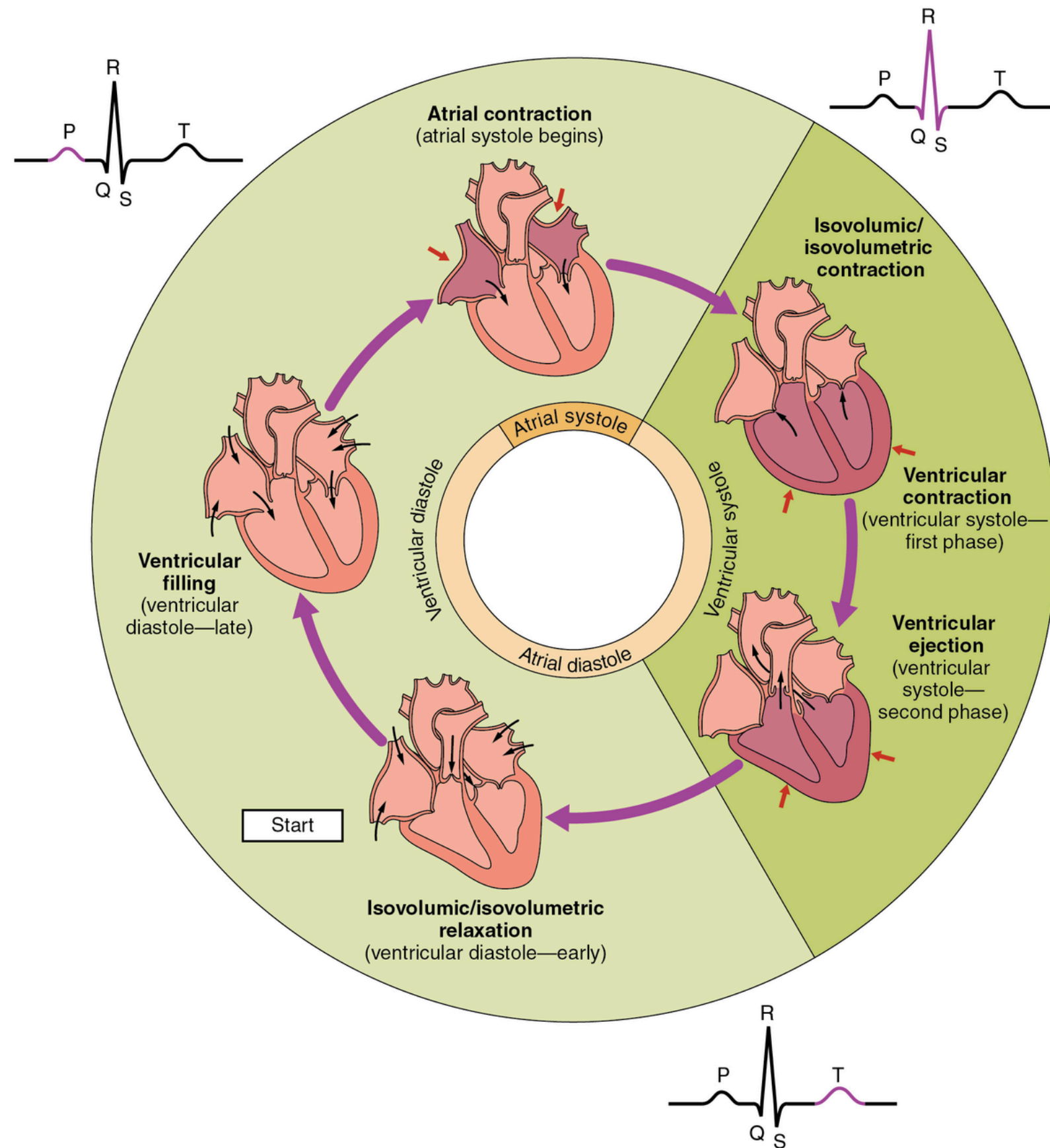


The cardiac cycle includes two parts,

1) During diastole: the **atria contract** and pump blood into the ventricles, while the ventricles relax and fill with blood from the atria.

2) During systole: the atria relax and collect blood from the lungs and body, while the **ventricles contract** and pump blood out of the heart.

1. Sequence of electrical activity and mechanical activity in a cardiac cycle



- At a normal heart rate of 75 beats per minute, one cardiac cycle lasts 0.8 second.
- Under resting conditions, systole occupies $\frac{1}{3}$ and **diastole $\frac{2}{3}$ of the cardiac cycle duration.**

Phase I: Atrial Systole “The final push of blood into the ventricles”

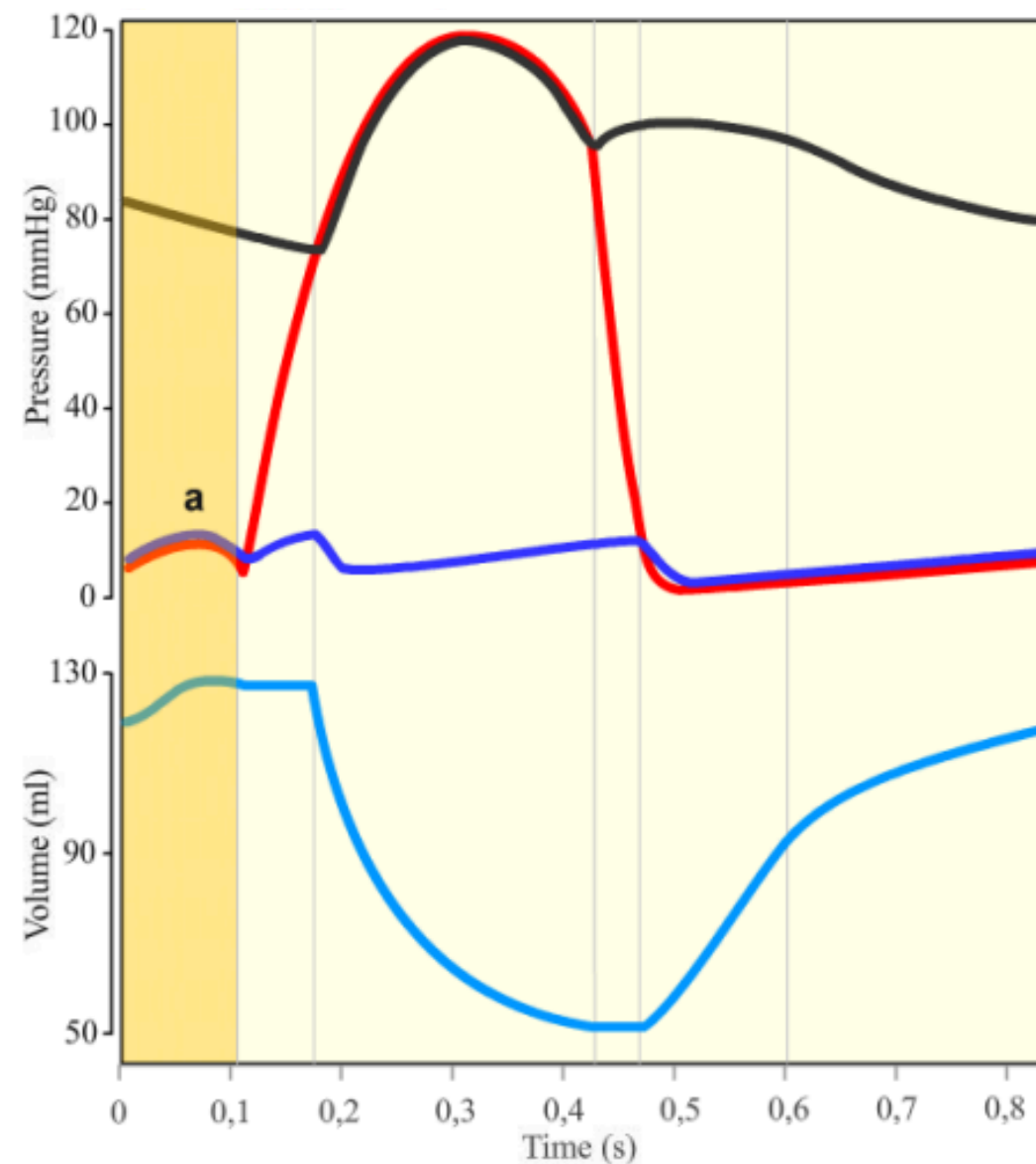
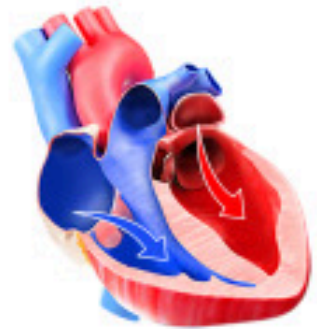


Figure 6.2. Atrial systole - pressure and volume changes

Red line - pressure in the left ventricle, black - the aortic pressure, dark blue - the pressure in the right atrium. light blue - the ventricular volume.

♥ Ventricles

- Atrial contraction pushes the final ~25% of ventricular filling into the ventricles.
 - Ventricular pressure increases only slightly.
 - At the end of atrial systole: Ventricular volume = ~130 mL called End-Diastolic Volume (EDV).

♥ Atria

- Atrial contraction → atrial pressure ↑
 - Produces the **a wave** in the venous pulse.

♥ Arteries

- Arterial pressure continues to gradually decrease during this phase.

ECG

- **P wave → atrial depolarization.**
 - Atrial systole begins around the end of the P wave.
 - PR segment → impulse travels through the AV node toward the ventricles.

Heart sounds

- **S4 (fourth heart sound)** may occur when **atrial contraction forces blood into a stiff** or poorly compliant ventricle.
 - May be associated with ventricular hypertrophy or reduced ventricular compliance.

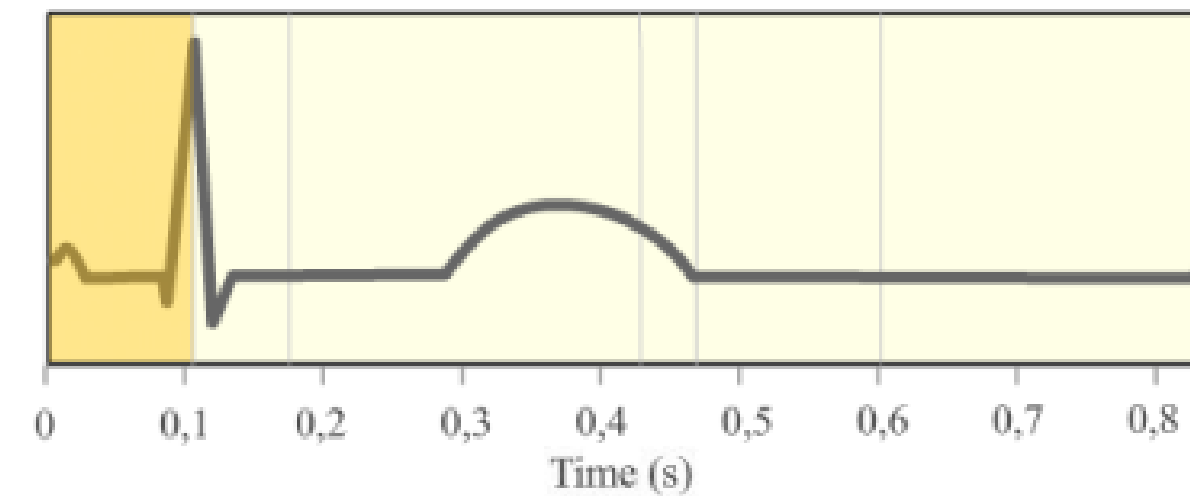


Fig. 6.3. Atrial systole - ECG

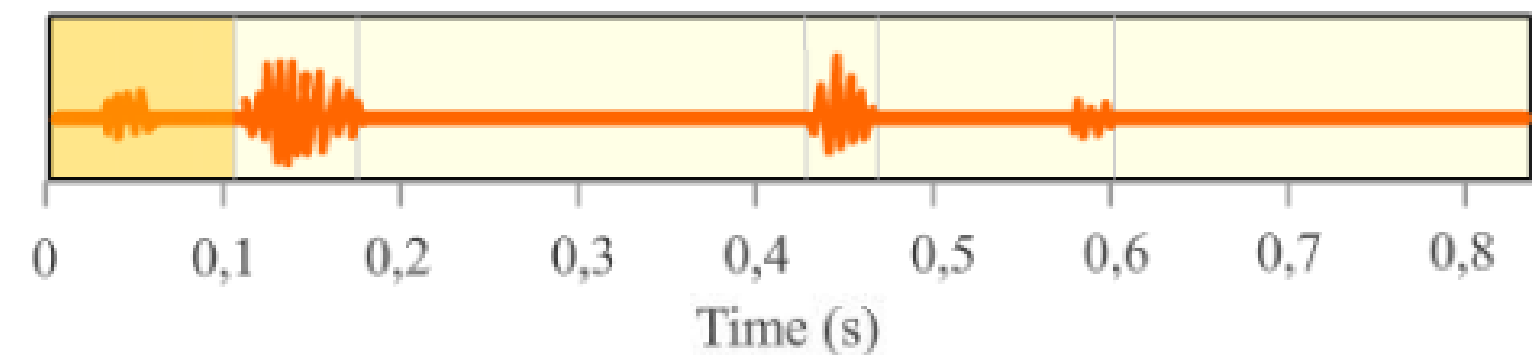


Fig. 6.4. Atrial systole – heart sounds

Phase II: Isovolumic Contraction “The ventricles contract—but no blood leaves yet”

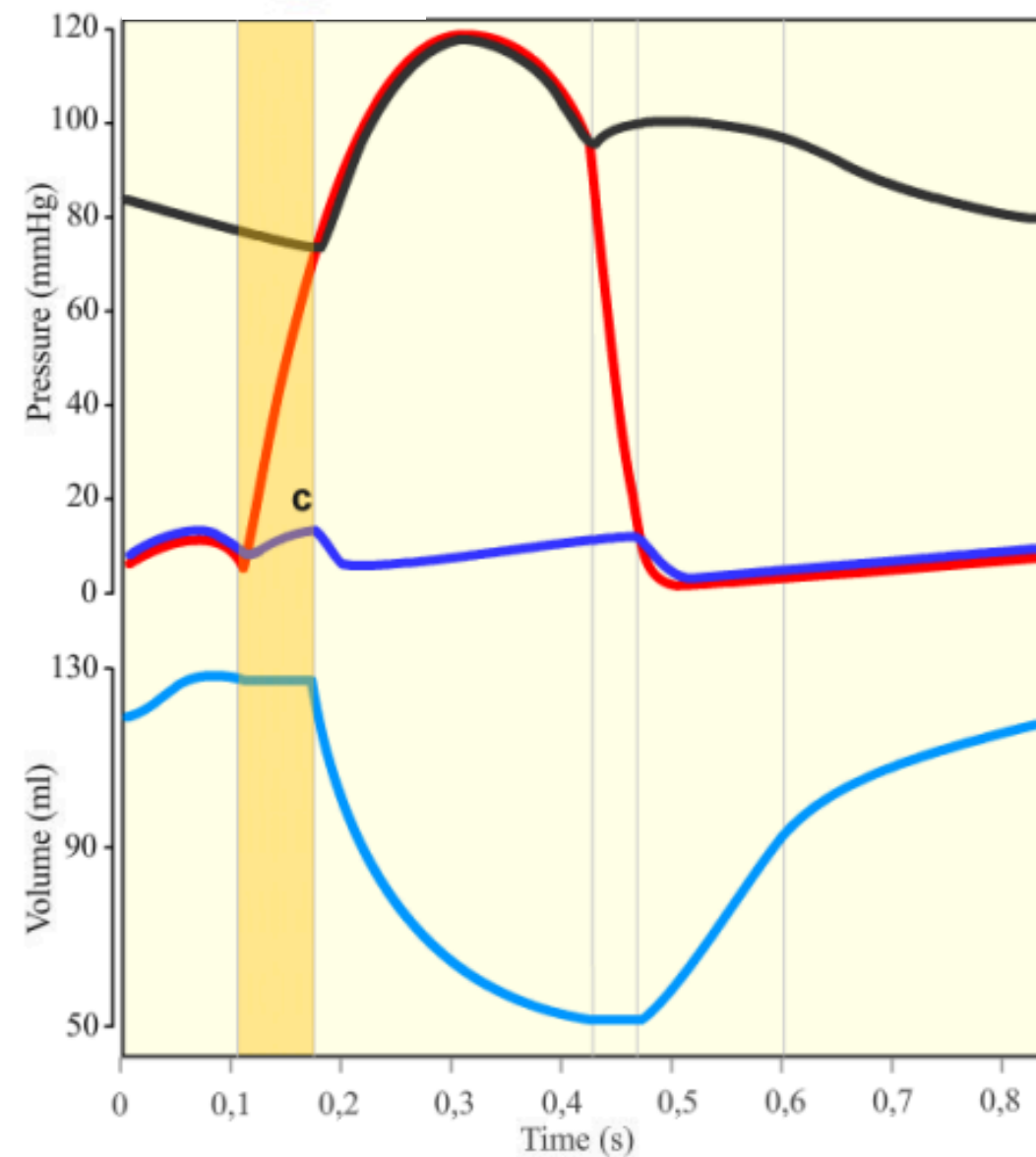


Figure 1.2. Isovolumic contraction - pressure and volume changes
Red line - pressure in the left ventricle, black - the aortic pressure, dark blue - the pressure in the right atrium, light blue - the ventricular volume.

♥ Ventricles

- Ventricular contraction begins → **ventricular pressure** ↑ rapidly
 - All valves are closed → no blood enters or leaves.
 - Ventricular volume remains constant at EDV ≈ 130 mL.

♥ Atria

- Rising ventricular pressure pushes the AV valves backward toward the atria.
 - Produces the **c wave** in the venous pulse.

💧 Arteries

- Arterial pressure continues to gradually decrease because blood continues flowing into the peripheral circulation.

ECG

- QRS complex → ventricular depolarization → initiates ventricular contraction.
 - Atrial repolarization produces the Ta wave normally hidden within the QRS complex.

Heart sounds

- **S1 (first heart sound)** → closure of the mitral and tricuspid valves.

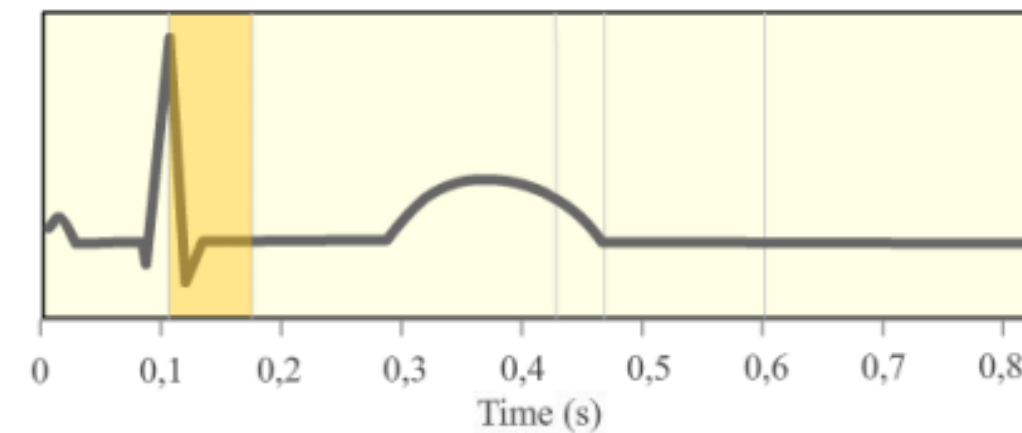


Fig. 1.3. Isovolumic contraction – ECG

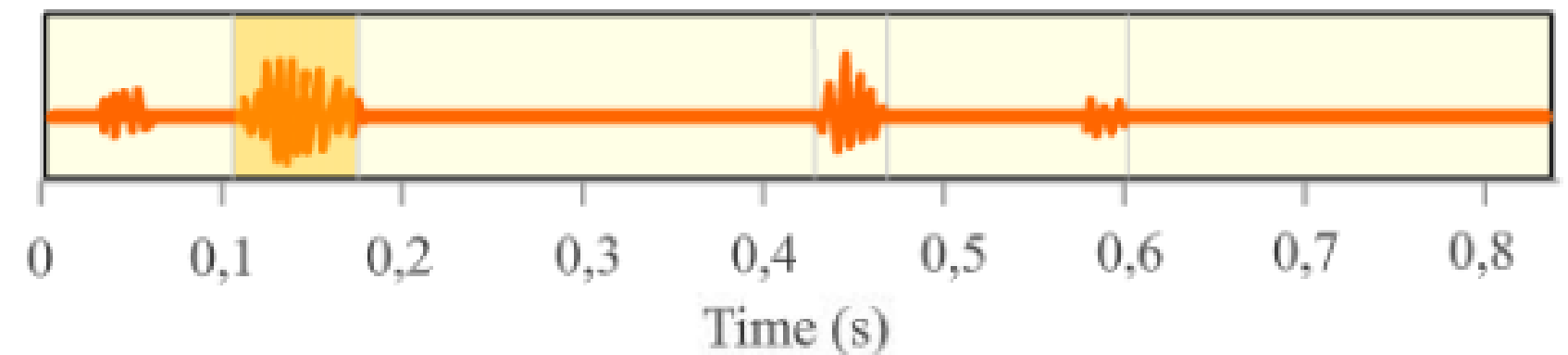


Fig. 1.4. Isovolumic contraction – heart sounds

Phase III: Ventricular Ejection “The semilunar valves open → blood leaves the ventricles”

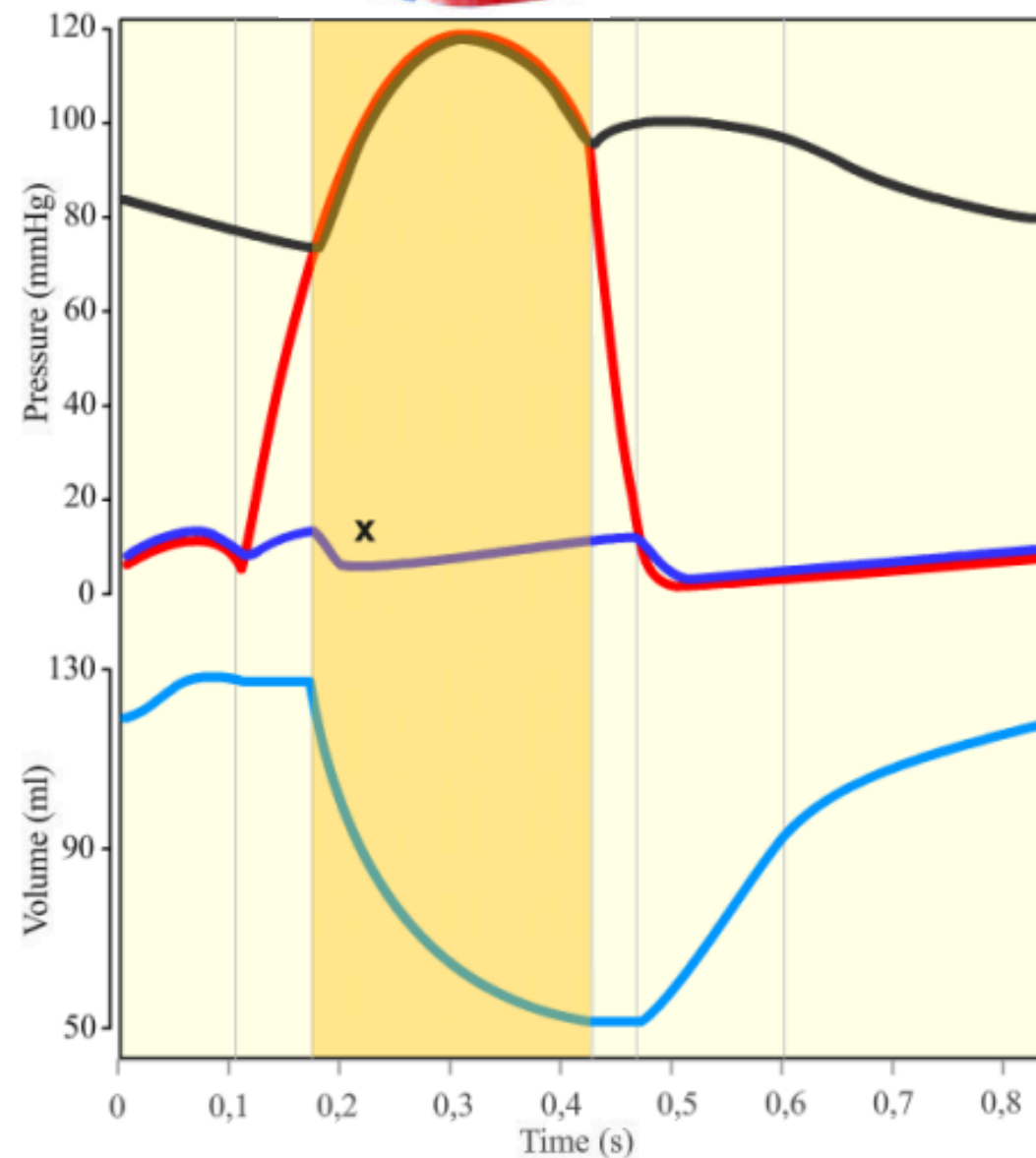


Figure 2.2. Ejection - pressure and volume changes
Red line - pressure in the left ventricle, black - the aortic pressure, dark blue - the pressure in the right atrium, light blue - the ventricular volume.

♥ Ventricles

- Ventricular pressure becomes greater than arterial pressure → **aortic & pulmonary valves open.**
 - **Rapid ejection:** ventricular pressure rises → blood is rapidly ejected.
 - **Reduced ejection:** ventricular pressure begins to fall → ejection slows.
 - Ventricular volume decreases to ≈ 60 mL called **End-systolic volume (ESV)**.
 - **Stroke volume (SV)** ≈ 70 mL is ejected from each ventricle (EDV-ESV).

♥ Atria

- Ventricular contraction pulls the AV valve plane downward → atrial pressure falls → produces the **x descent (x wave)** in the venous pulse.
 - **At the same time, the atria are filling** with blood for the next cycle.

♥ Arteries

- Rapid ejection → arterial pressure rises to systolic pressure:
 - Aorta ≈ 120 mmHg
 - Pulmonary artery ≈ 25 mmHg
- During reduced ejection → arterial pressure gradually falls.



ECG

- **ST segment:** ventricles are completely depolarized.
- T wave: ventricular repolarization begins during the second half of ejection.

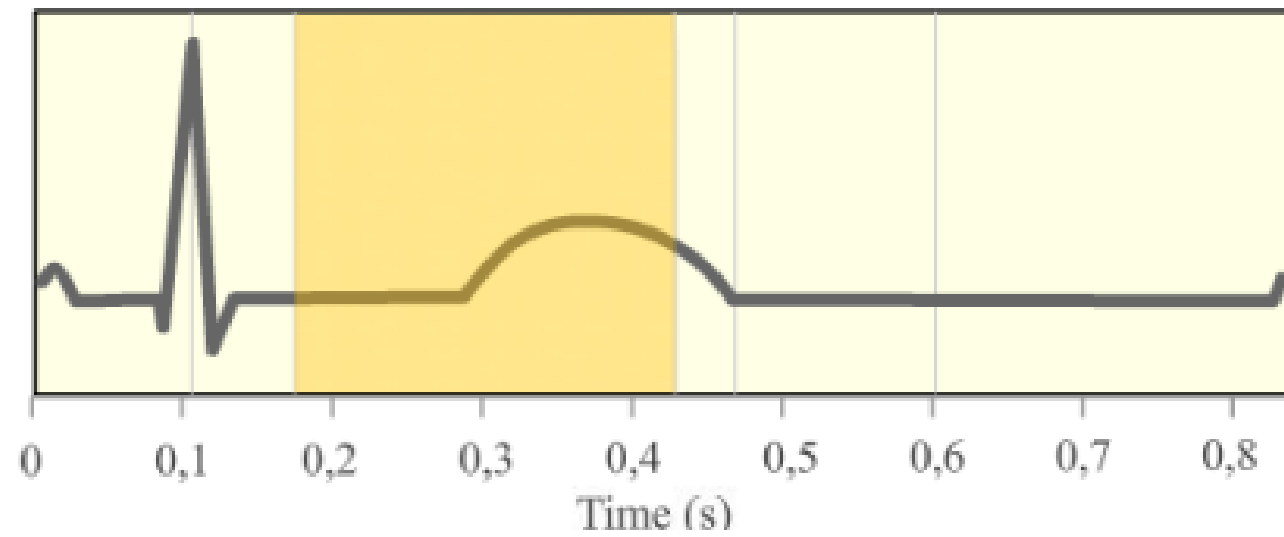


Fig 2.3. Ejection - ECG

Phase IV: Isovolumic Relaxation “The ventricles relax—but no blood enters yet”

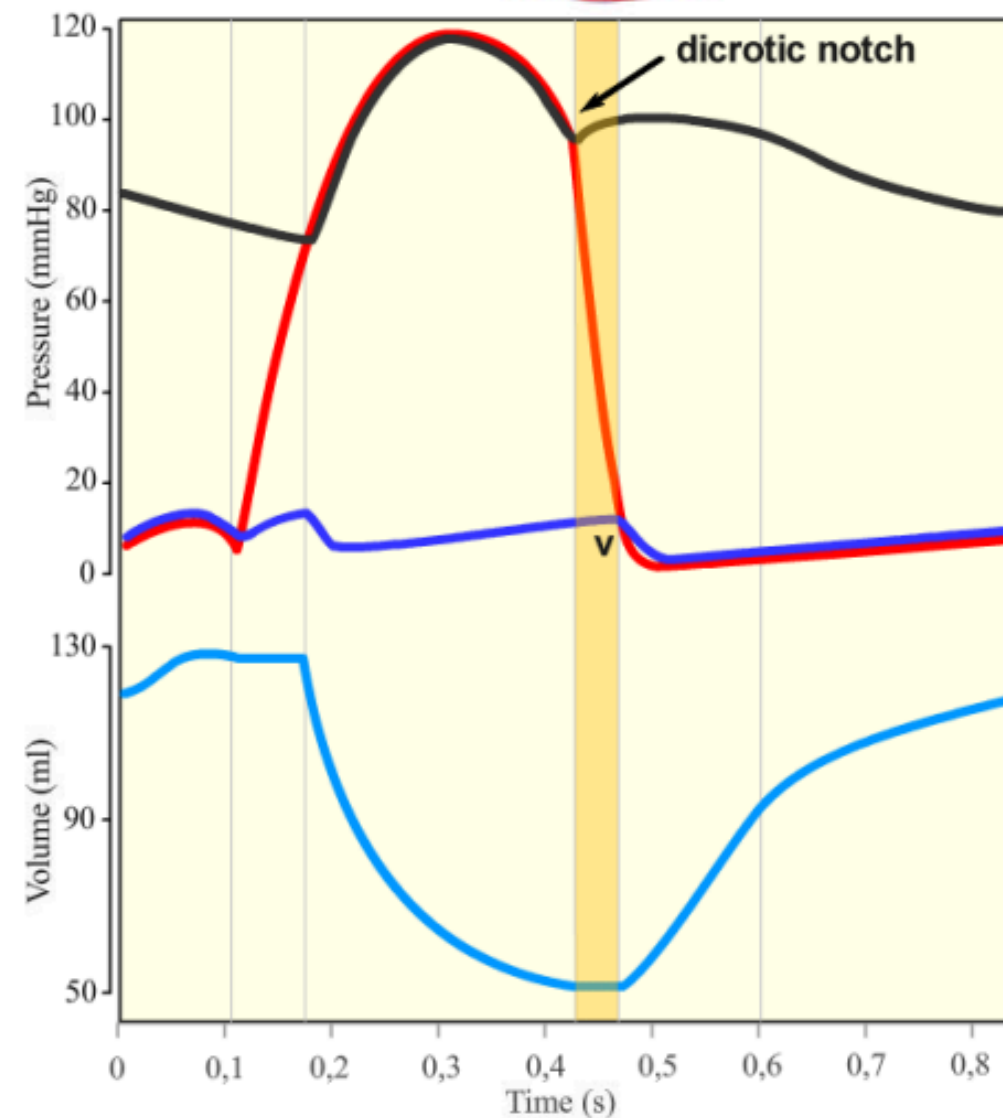


Figure 3.2. Isovolumic relaxation - pressure and volume changes
Red line - pressure in the left ventricle, black - the aortic pressure, dark blue - the pressure in the right atrium, light blue - the ventricular volume.

♥ Ventricles

- Ventricular relaxation begins → **ventricular pressure falls rapidly**.
 - All valves are closed → no blood enters or leaves.
 - By the end of this phase, ventricular pressure approaches 0 mmHg.

♥ Atria

- Blood continues flowing from the veins → atria fill while AV valves remain closed.
- Atrial pressure gradually increases → produces the **v wave** in the venous pulse.

♥ Arteries

- Closure of the aortic and pulmonary valves causes a brief interruption/reversal of flow--- **dicrotic notch** in the aortic pressure waveform.
 - Arterial pressure then continues to decline during diastole.

ECG

- T wave ends → ventricular repolarization is completed.

Heart sounds

- **S2 (second heart sound)** → closure of the aortic and pulmonary valves.

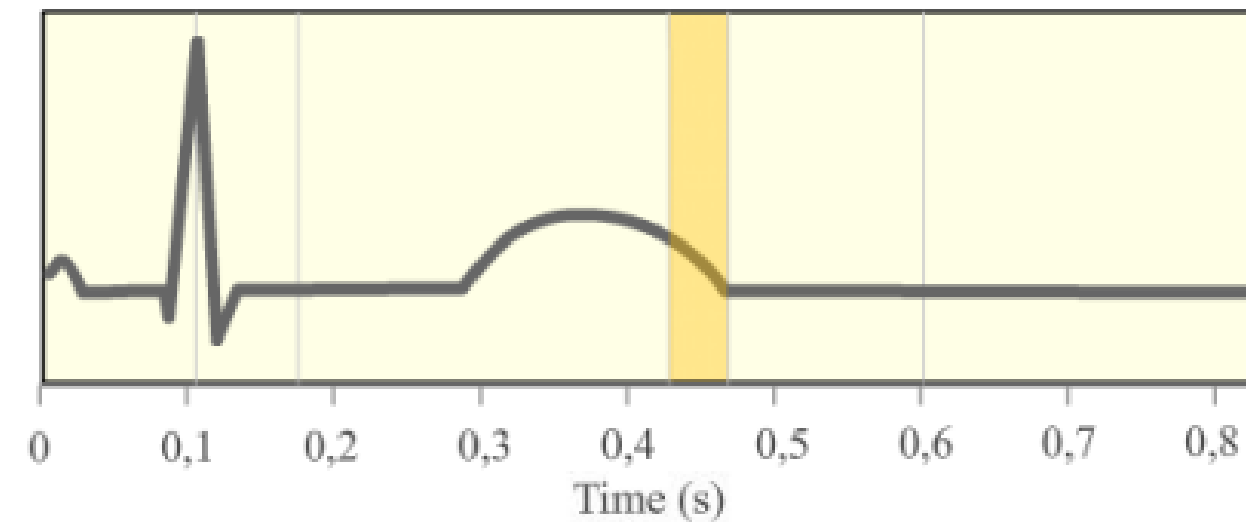


Fig. 3.3. Isovolumic relaxation - ECG

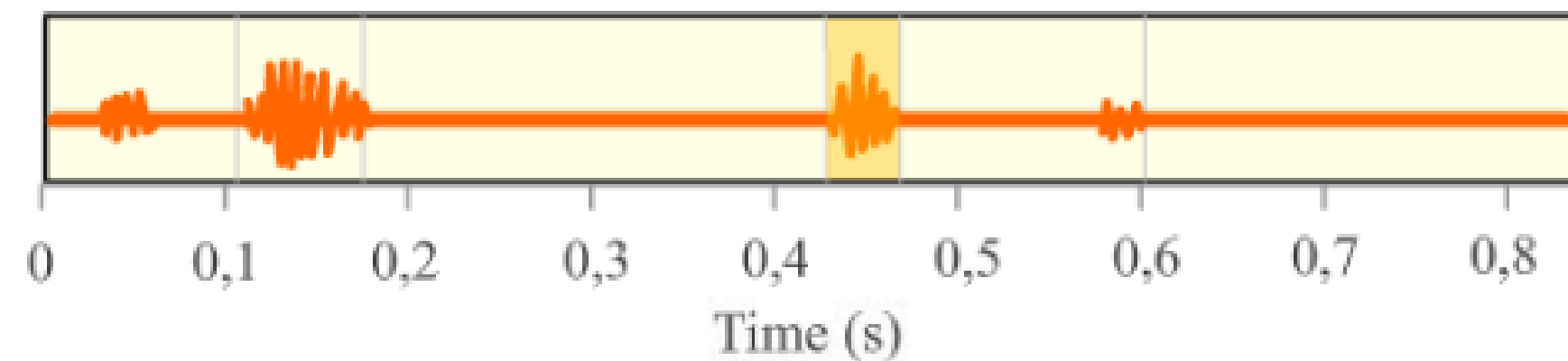


Fig. 3.4. Isovolumic relaxation – heart sounds

Phase V: Rapid Ventricular Filling “Blood rapidly flows into relaxed ventricles”



♥ Ventricles

- Ventricular pressure falls below atrial pressure → **AV valves open**.
 - Blood accumulated in the atria rapidly flows into the ventricles.
 - Ventricular volume ↑ rapidly, while **ventricular pressure changes very little because the ventricles are relaxed** (passive ventricular filling).

♥ Atria

- Blood rapidly leaves the atria → atrial pressure ↓.
 - Produces the **y descent** in the venous pulse.

♥ Arteries

- Semilunar valves remain closed.
- Arterial pressure gradually decreases during diastole.
 - Elastic recoil of large arteries prevents arterial pressure from falling to zero.
- Minimum arterial pressure = diastolic pressure:
 - Systemic ≈ 80 mmHg
 - Pulmonary ≈ 8 mmHg

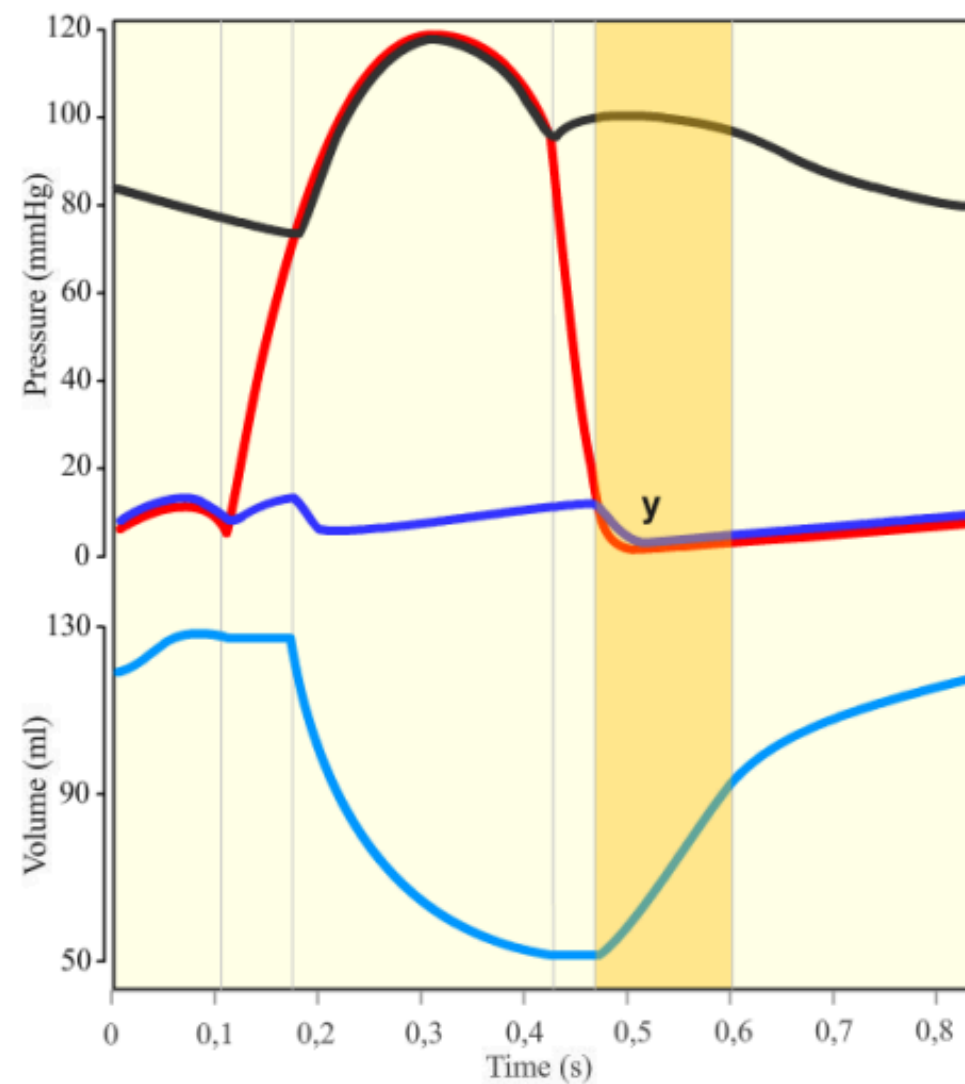


Figure 4.2. Rapid ventricular filling - pressure and volume changes
Red line - pressure in the left ventricle, black - the aortic pressure, dark blue - the pressure in the right atrium, light blue - the ventricular volume.

ECG

- No major electrical activity is detected during this phase.
 - ECG shows an isoelectric (flat) line.

Heart sounds

- **S3 (third heart sound)** may occur during rapid ventricular filling.
 - S3 can be normal in children and young adults, but in older adults it **may indicate volume overload or ventricular dysfunction.**

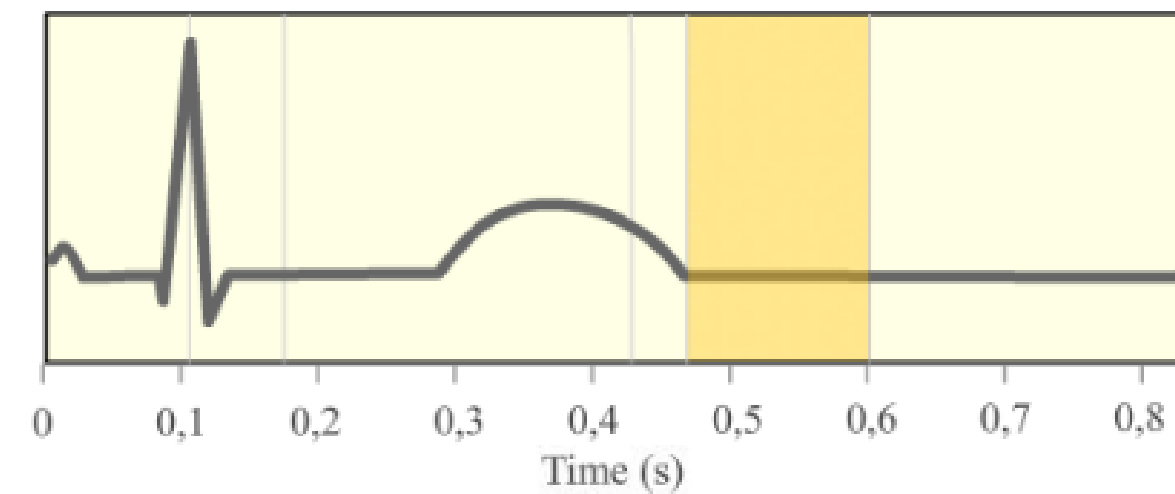


Fig. 4.3. Rapid ventricular filling - ECG

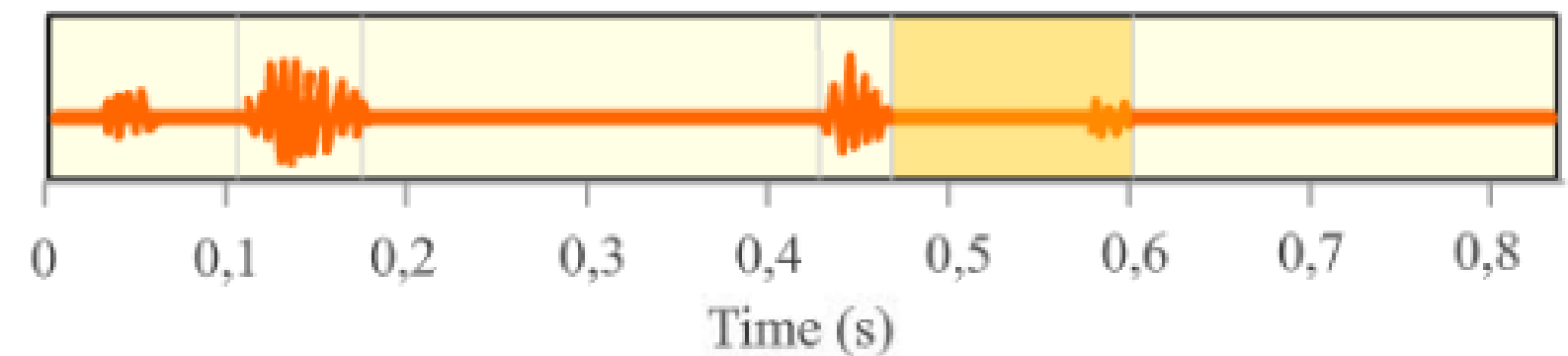


Fig. 4.4. Rapid ventricular filling – heart sounds

Phase VI: Slow Ventricular Filling (Diastasis) “Ventricles continue to fill slowly”

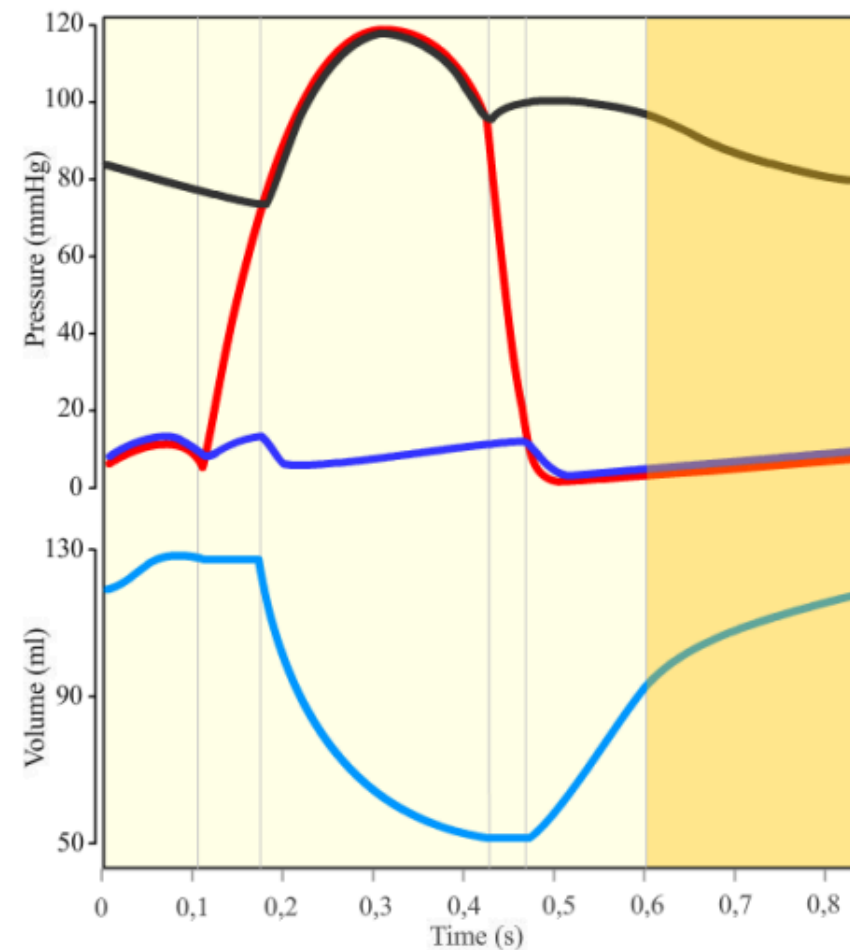


Figure 5.1. Slow ventricular filling - pressure and volume changes
Red line - pressure in the left ventricle, black - the aortic pressure, dark blue - the pressure in the right atrium. Light blue - the ventricular volume.

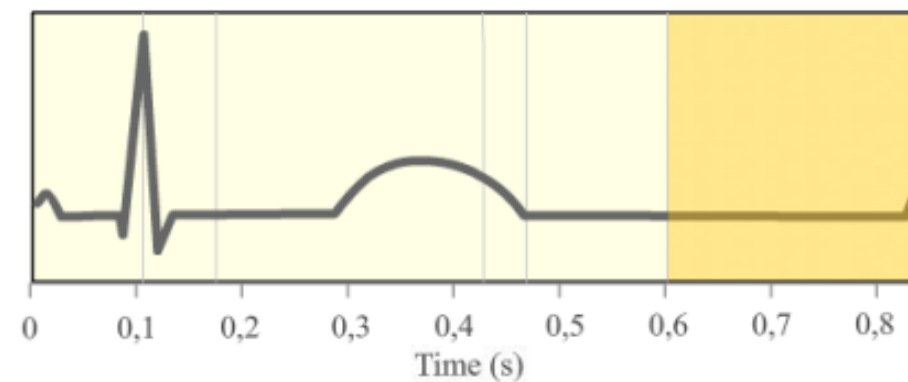


Fig. 5.2. Slow ventricular filling – ECG

♥ Ventricles

- Blood continues to flow from the atria → ventricles, but at a slow rate.
 - Ventricular volume gradually increases.
 - Ventricular pressure remains very low, close to 0 mmHg.

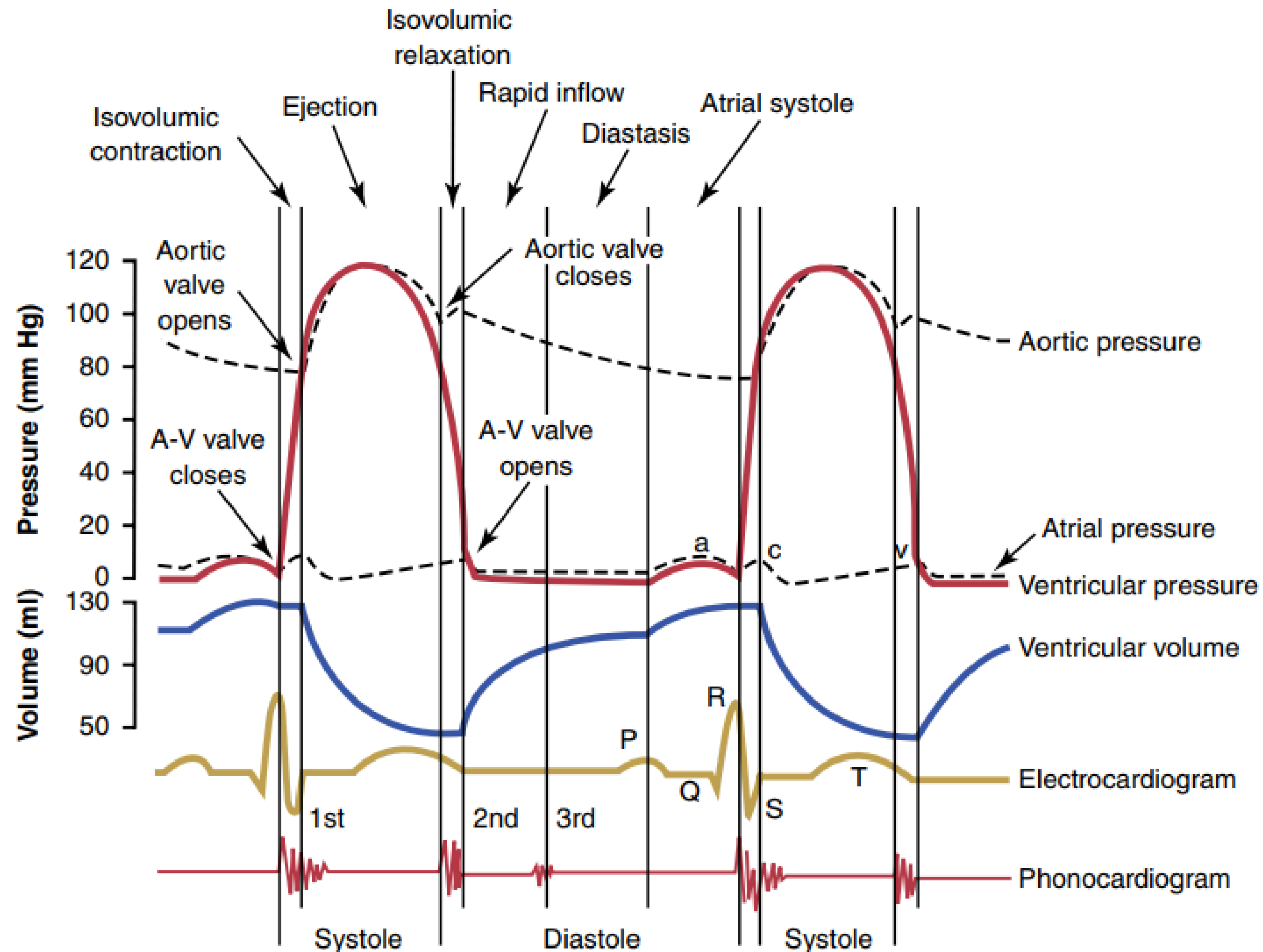
💧 Arteries

- Arterial pressure continues to gradually decrease in both systemic and pulmonary circulations.

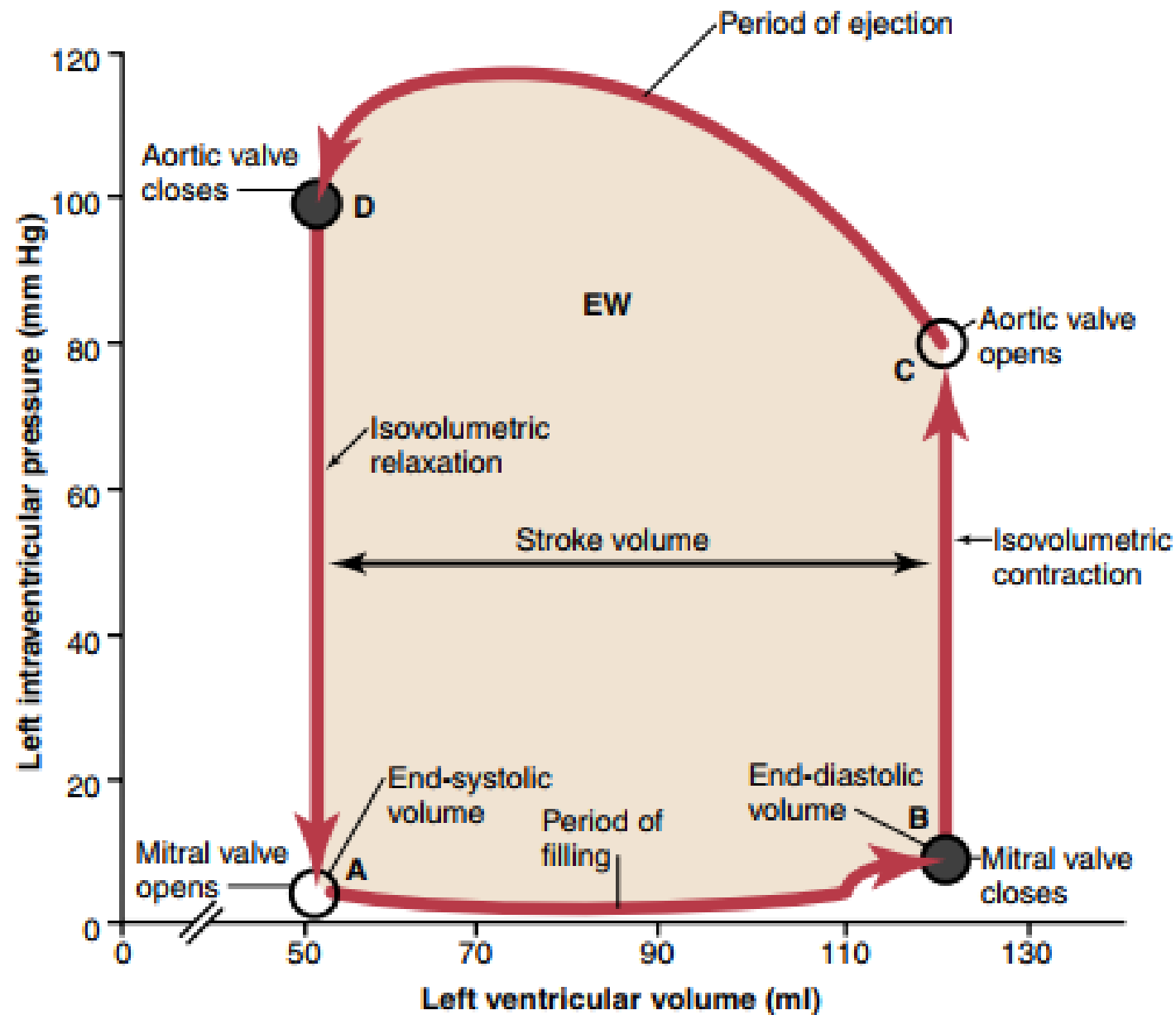
📊 ECG

- SA node generates the next impulse.
 - Depolarization spreads across the atria → P wave begins.
 - The P wave signals the transition toward the next atrial systole.

2. Pressures, volume and flow in the heart and great vessels in a cardiac cycle



Pressure-Volume Relationship



- **Phase I: Period of Filling.**
 - Volume ↑ | Pressure slightly ↑
- **Phase II: Period of Isovolumic Contraction.**
 - Volume ↔ | Pressure ↑↑
- **Phase III: Period of Ejection.**
 - Volume ↓ | Pressure ↑ then ↓
- **Phase IV: Period of Isovolumic Relaxation.**
 - Volume ↔ | Pressure ↓↓

Systolic Dysfunction

“The ventricle cannot contract strongly enough to eject blood.”

❤️ ↓ **Contractility** → cardiac muscle cannot shorten effectively against the afterload.

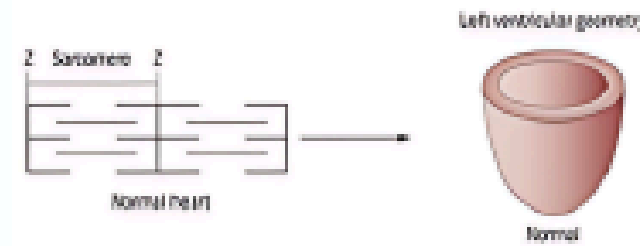
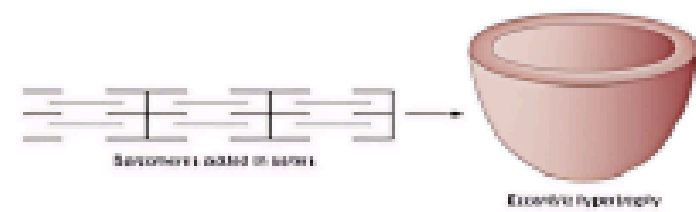
- The LV has difficulty generating enough pressure to eject blood into the aorta.
 - ↓ Stroke volume (SV)
 - ↑ End-systolic volume (ESV).
- Over time, the LV may undergo eccentric remodeling/dilation → ↑ **LV diastolic volume (EDV)**.

📊 Pressure–Volume Loop

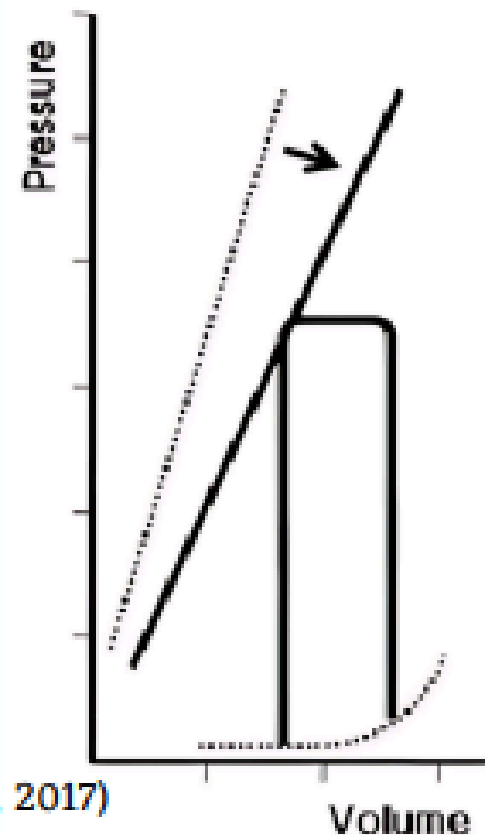
- Reduced contractility → the end-systolic pressure–volume relationship (**ESPVR**) becomes **less steep**.
 - **ESV increases** → **SV decreases**

👨‍⚕️ Common causes

- **Myocardial infarction/Chronic hypertension/Dilated cardiomyopathy/Some arrhythmias**

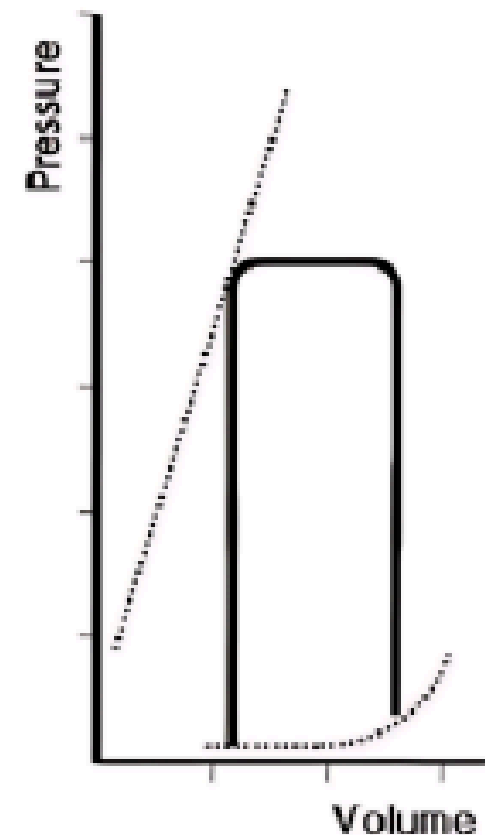


Systolic dysfunction



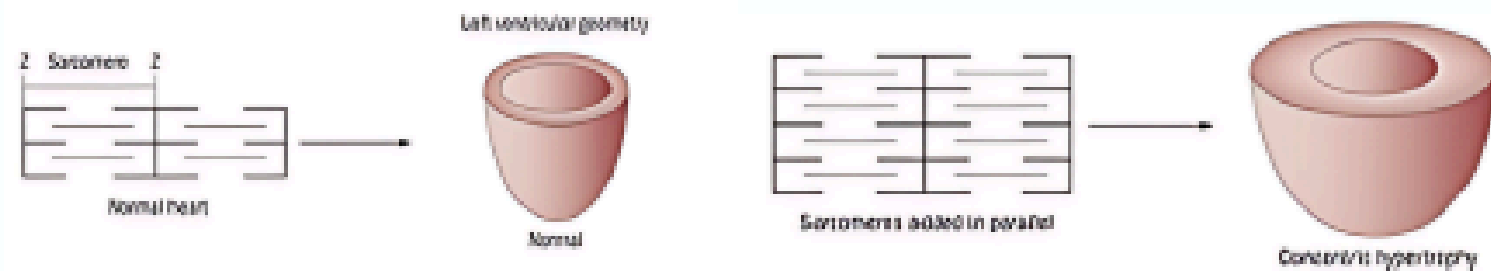
(Sebastian et al., 2017)

Normal



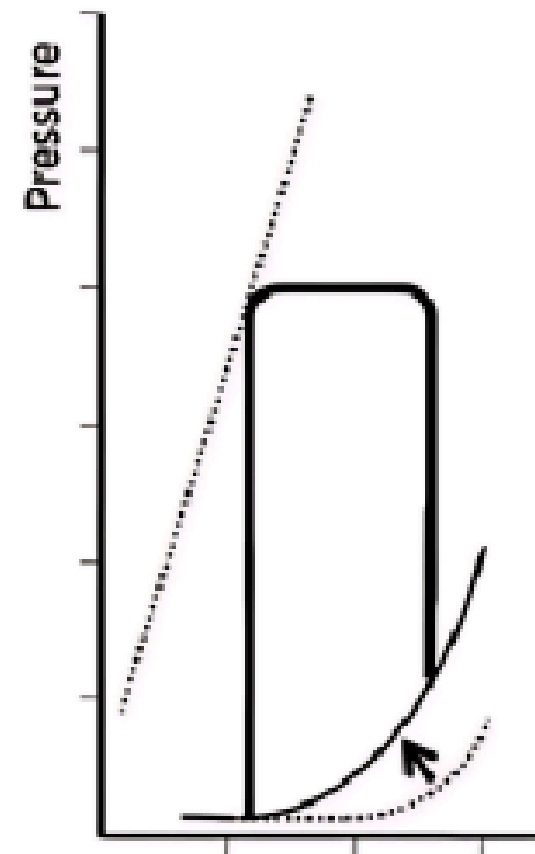
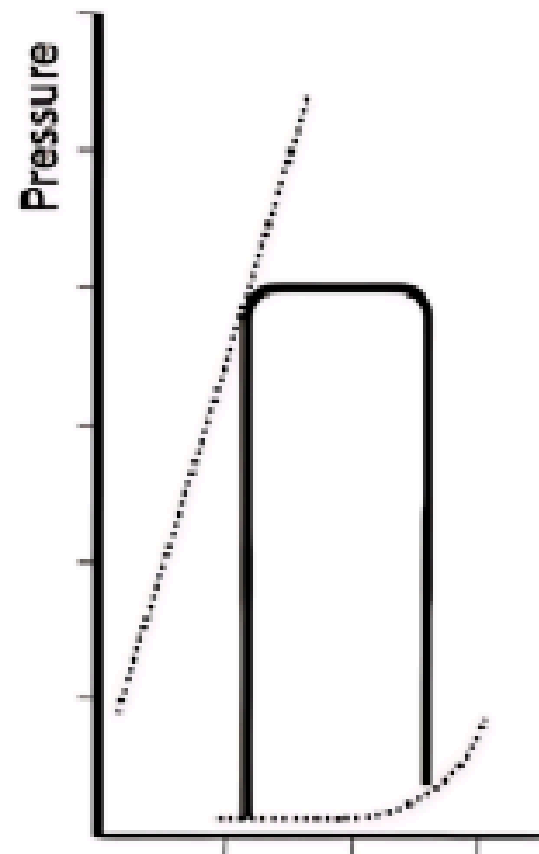
Diastolic Dysfunction

“The ventricle is too stiff or relaxes too slowly to fill normally.”



Normal

Diastolic dysfunction



(Sebastian et al., 2017)

♥ ↓ **Ventricular relaxation** → ↑ Ventricular stiffness → the LV cannot expand easily during filling → **the LV needs higher atrial pressure to fill adequately.**

- Blood may accumulate upstream → ↑ atrial/pulmonary venous pressure.
- Commonly associated with **concentric LV hypertrophy/remodeling** (ventricular wall becomes **thicker and less compliant**).

Pressure-Volume Loop

- A given ventricular volume requires higher pressure.
 - The diastolic pressure-volume relationship shifts upward/leftward.
 - LV diastolic volume may remain normal, despite the increased filling pressure.

Common causes

- Chronic hypertension/Ischemic heart disease/Hypertrophic cardiomyopathy

3. Heart sound and murmurs

1. Valves open and close passively

- Open: upstream pressure > downstream pressure
- Close: downstream pressure > upstream pressure

2. Abnormal valve function

- **Regurgitation:** valve does not close completely → backward blood flow
- **Stenosis:** valve opening becomes narrow → restricted forward blood flow

3. Murmur

- Abnormal/turbulent blood flow through a stenotic or regurgitant valve
- Produces characteristic heart sounds (“murmurs”) during specific phases of the cardiac cycle

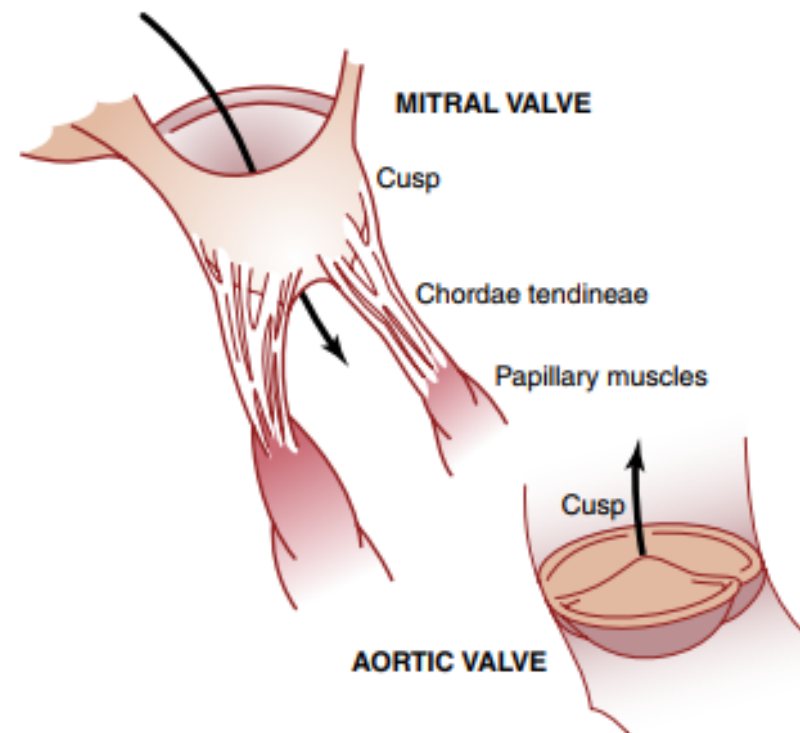
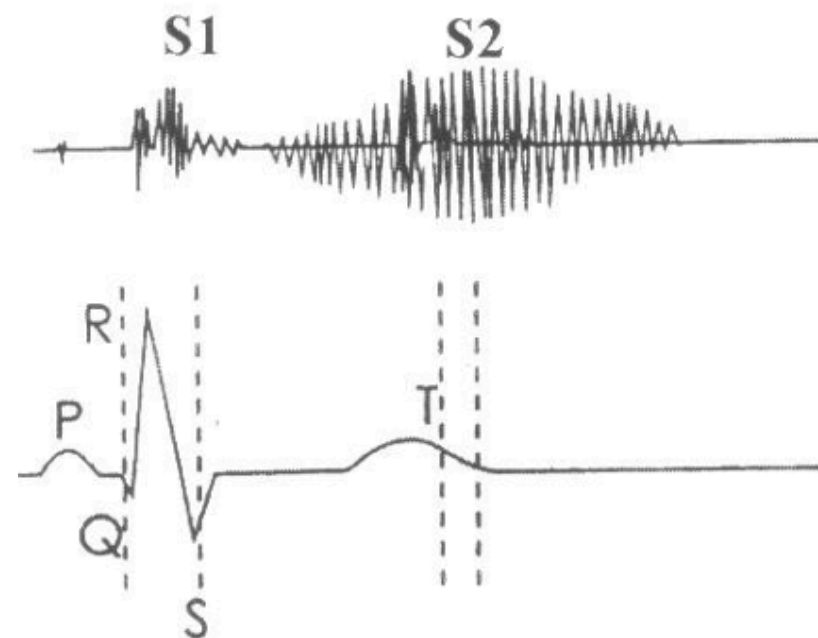
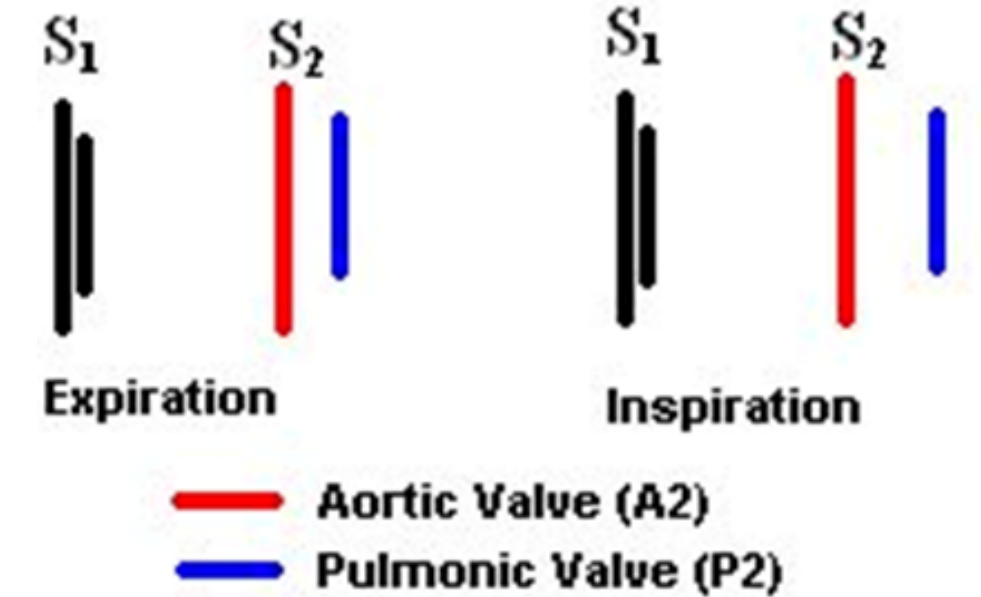
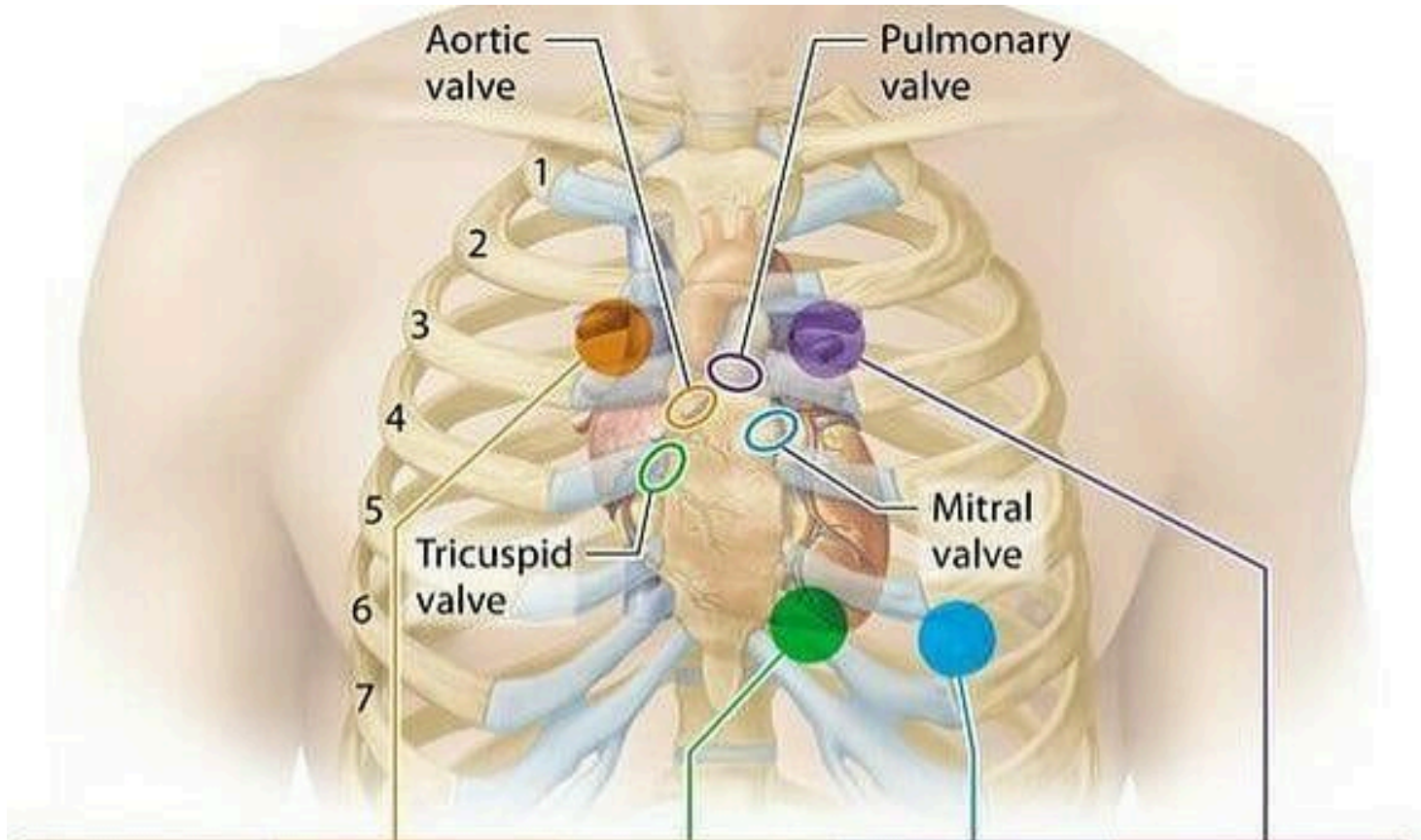


Figure 9-9. Mitral and aortic valves (the left ventricular valves).

Widened split S₂



- **First sound (S₁)**
 - at the isovolumic contraction
 - **AV valves close**
- **Second sound (S₂)**
 - at the isovolumic relaxation
 - **semilunar valves close**
- **Third sound (S₃)**
 - at the period of rapid ventricular filling
 - inrush of blood
- **Fourth sound (S₄)**
 - high atrial pressure or ventricular hypertrophy
 - rare case in normal adults



AREA	Aortic	Tricuspid	Mitral	Pulmonary
LOCATION OF SOUND	Second intercostal space, right sternal border	Fifth intercostal space, left sternal border	Fifth intercostal space, mid-clavicular line	Second intercostal space, left sternal border
TIMING OF SOUND	Aortic valve is heard here during S2.	Tricuspid valve is heard here during S1.	Mitral valve is heard here during S1.	Pulmonary valve is heard here during S2.

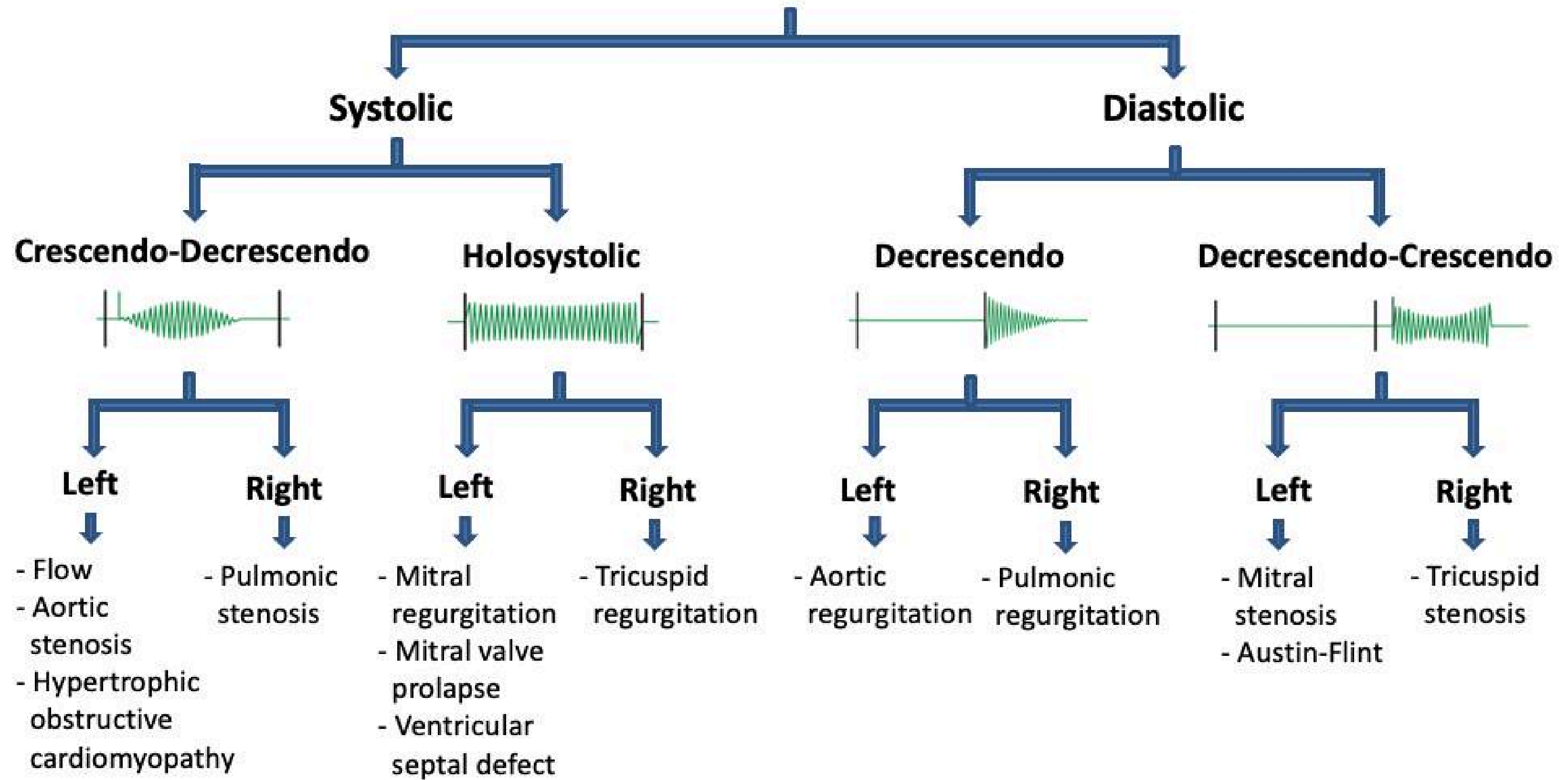
Heart Sounds and Murmurs

Watch on YouTube

Alila MEDICAL MEDIA

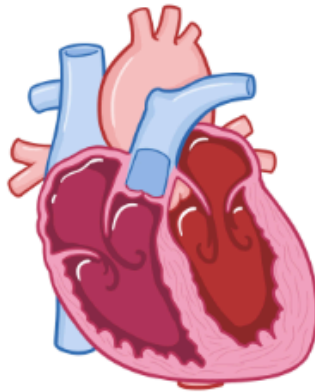
Frequency of murmurs depends on blood velocity
 Transmission of murmurs depend on direction of the blood flow

Heart Murmurs Adults



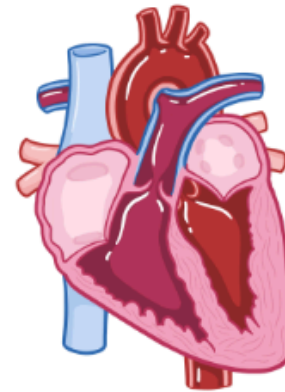
STROKE VOLUME, EJECTION FRACTION, & CARDIAC OUTPUT

DIASTOLE



EX. 120mL BLOOD
(END-DIASTOLIC VOLUME)

SYSTOLE



50mL BLOOD
(END-SYSTOLIC VOLUME)

$$S.V. = (END-DIASTOLIC VOL.) - (END-SYSTOLIC VOL.)$$

$$S.V. = 120mL - 50mL = 70mL$$

$$E.F. = \frac{S.V.}{(END-DIASTOLIC VOL.)}$$

$$E.F. = \frac{70mL}{120mL} = 58\%$$

S.V. - STROKE VOLUME
E.F. - EJECTION FRACTION
C.O. - CARDIAC OUTPUT

$$C.O. = S.V. \times \text{HEART RATE}$$

$$C.O. = 70mL/beat \times 70 \text{ beats/min} = 4.9L/min$$

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A 58-year-old man visits the cardiology clinic because of **increasing shortness of breath when climbing stairs.**

Echocardiography shows:

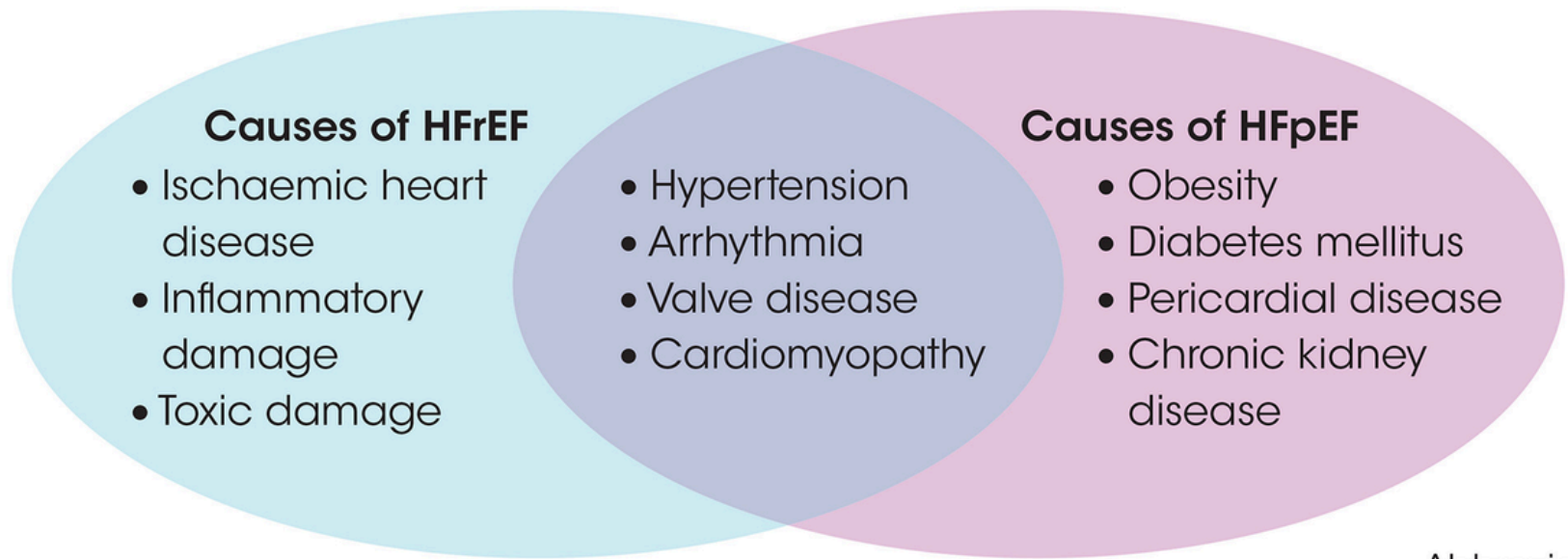
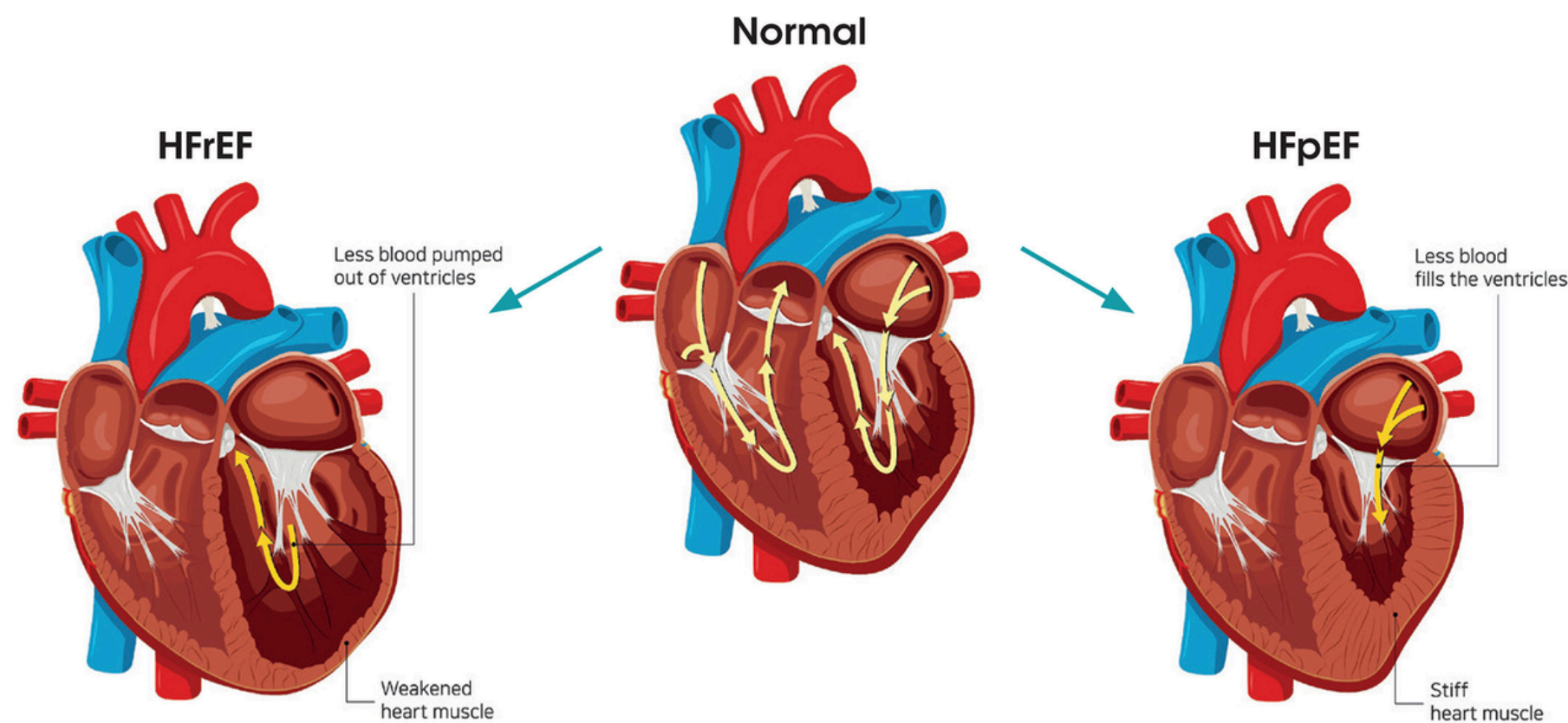
End-diastolic volume (EDV) = 150 mL

End-systolic volume (ESV) = 90 mL

Based on the EF, would you consider ventricular systolic function to be normal or reduced?



Ejection Fraction (EF) & Heart Failure

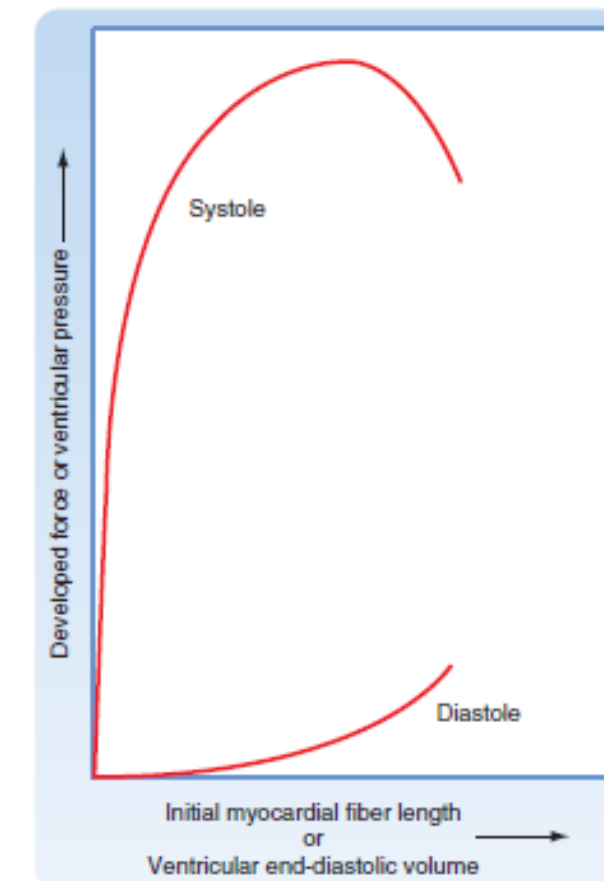
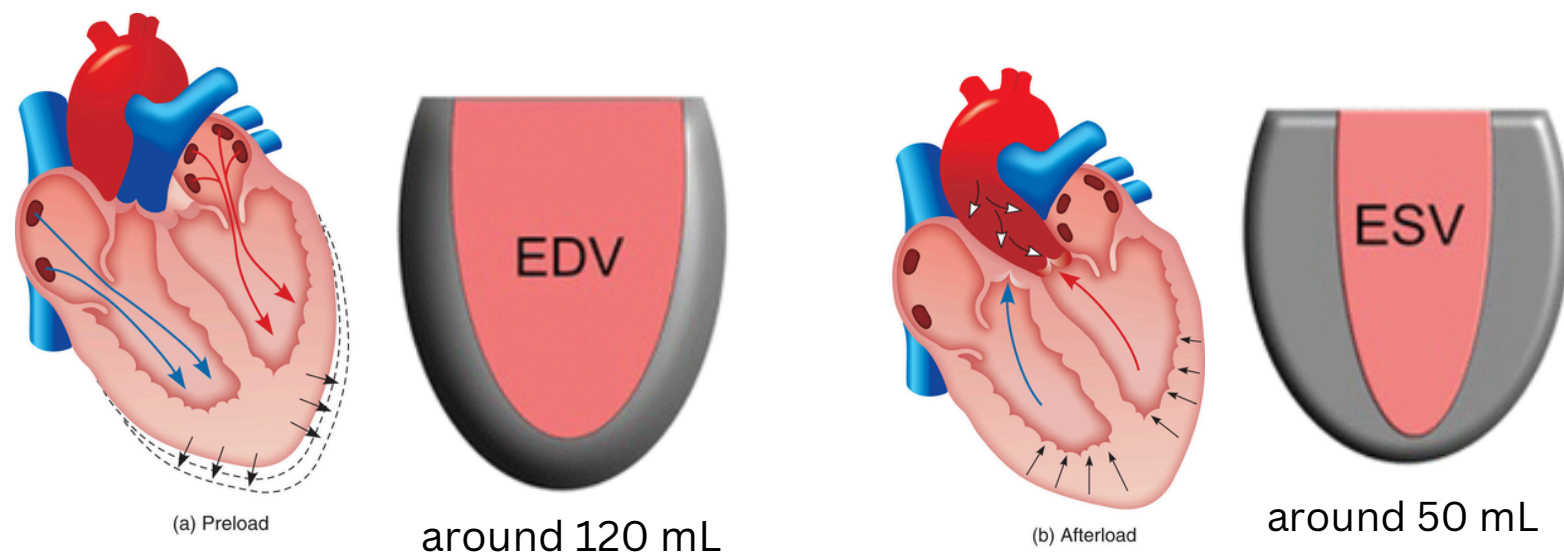


Abbreviations:
HFrEF: heart failure with reduced ejection fraction;
HFpEF: heart failure with preserved ejection fraction

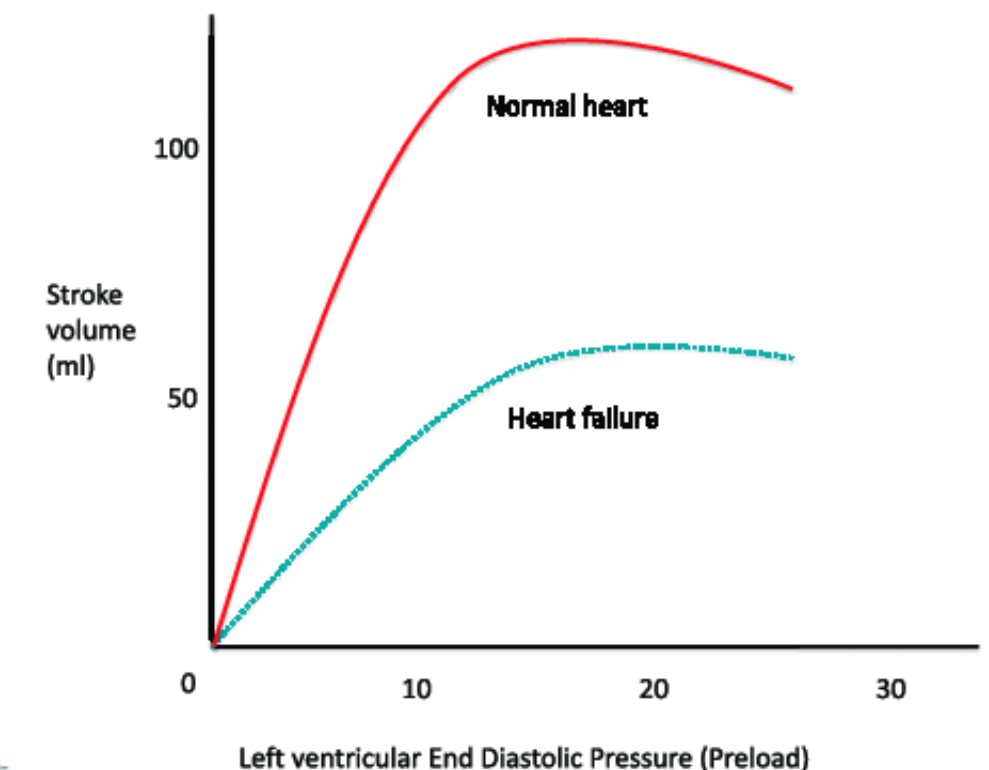
	HFrEF	HFpEF
EF	Reduced	Preserved
Main problem	Poor contraction	Poor relaxation/filling
Systole	↓ Contractility	Relatively preserved
Diastole	May be impaired	↓ Compliance / impaired relaxation
Blood ejected	↓	May be relatively normal %
Key concept	“Cannot pump well”	“Cannot fill well”

The concepts of preload and afterload applied to the contraction of the heart?

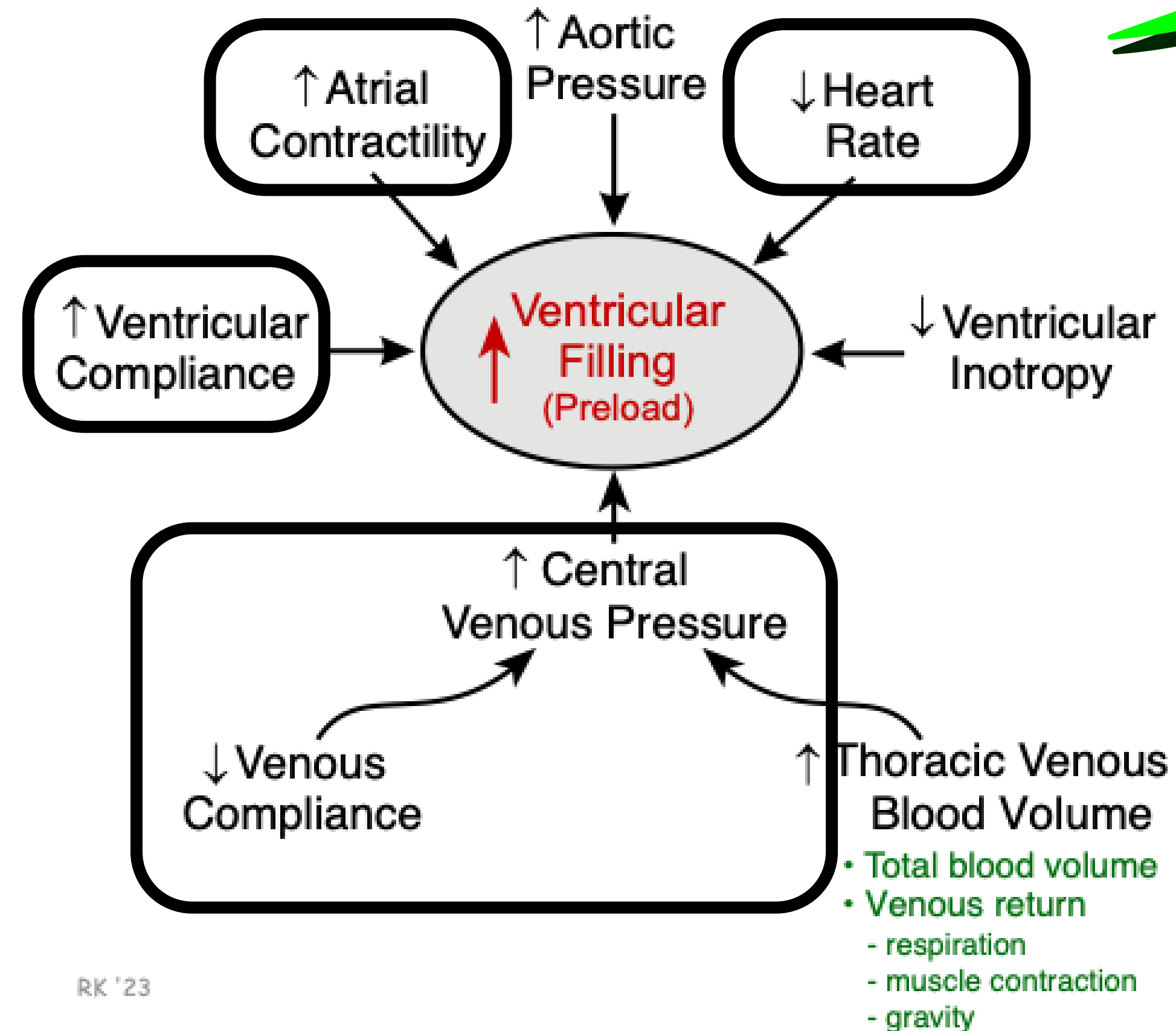
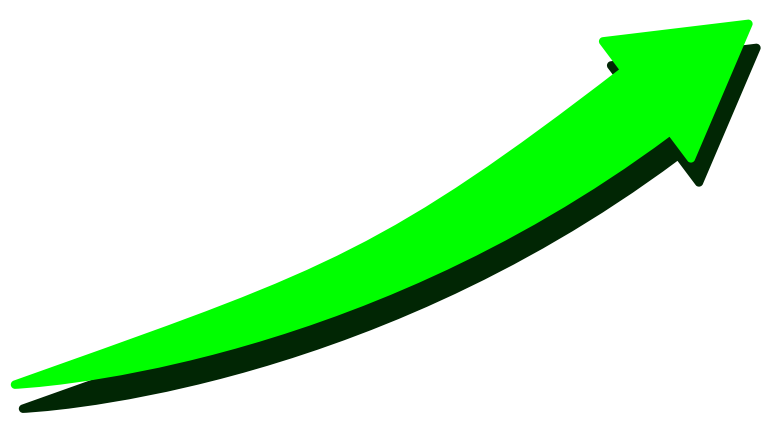
- **End diastolic volume (EDV)** is the blood volume in left ventricle at the end of filling.
 - **Preload** is the force that stretches the relaxed muscle fibers, thereby increasing their resting length.
 - **The Frank-Starling mechanism** (Starling's Law of the Heart) is the heart's **intrinsic ability** to adapt its output to changes in blood flow (venous return) by **increasing the force of contraction when more blood fills the heart**.
- **End systolic volume (ESV)** is the blood volume in left ventricle at the end of contraction.
 - **Afterload** is the force added to the muscle (against which the contracting muscle must act)



• Fig. 16.36 Relationship of myocardial resting fiber length (sarcomere length) or end-diastolic volume to the force developed or peak systolic ventricular pressure during ventricular contraction in an intact heart. (Redrawn from Patterson SW, et al. *J Physiol.* 1914;48:465.)

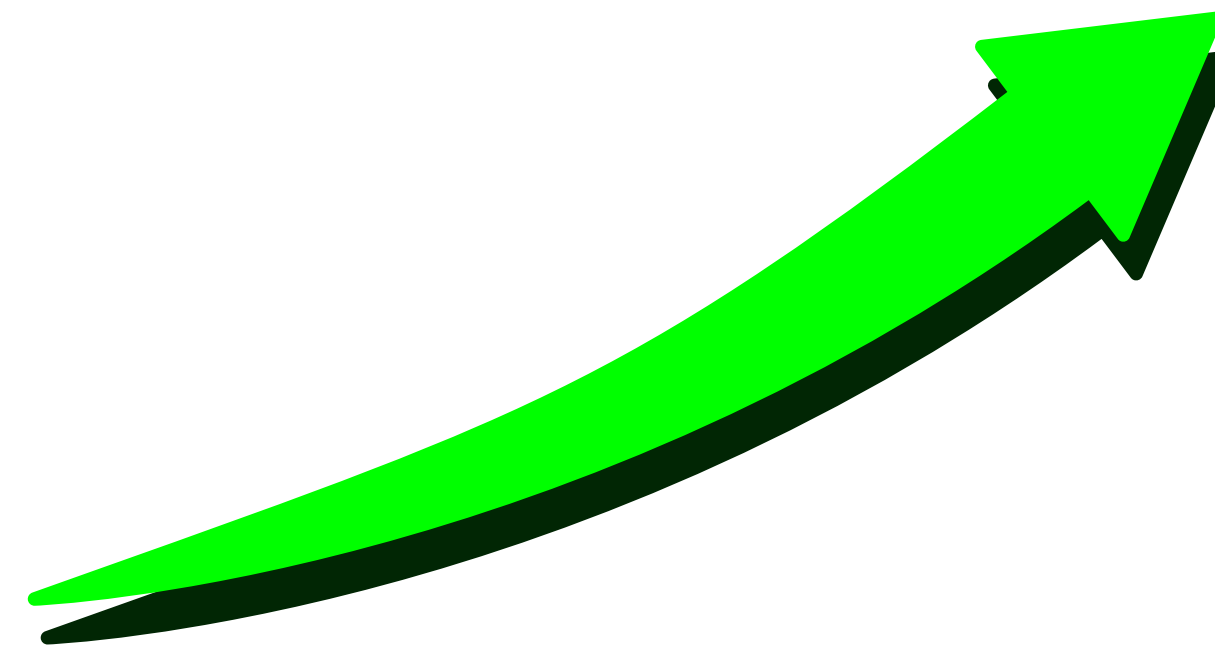
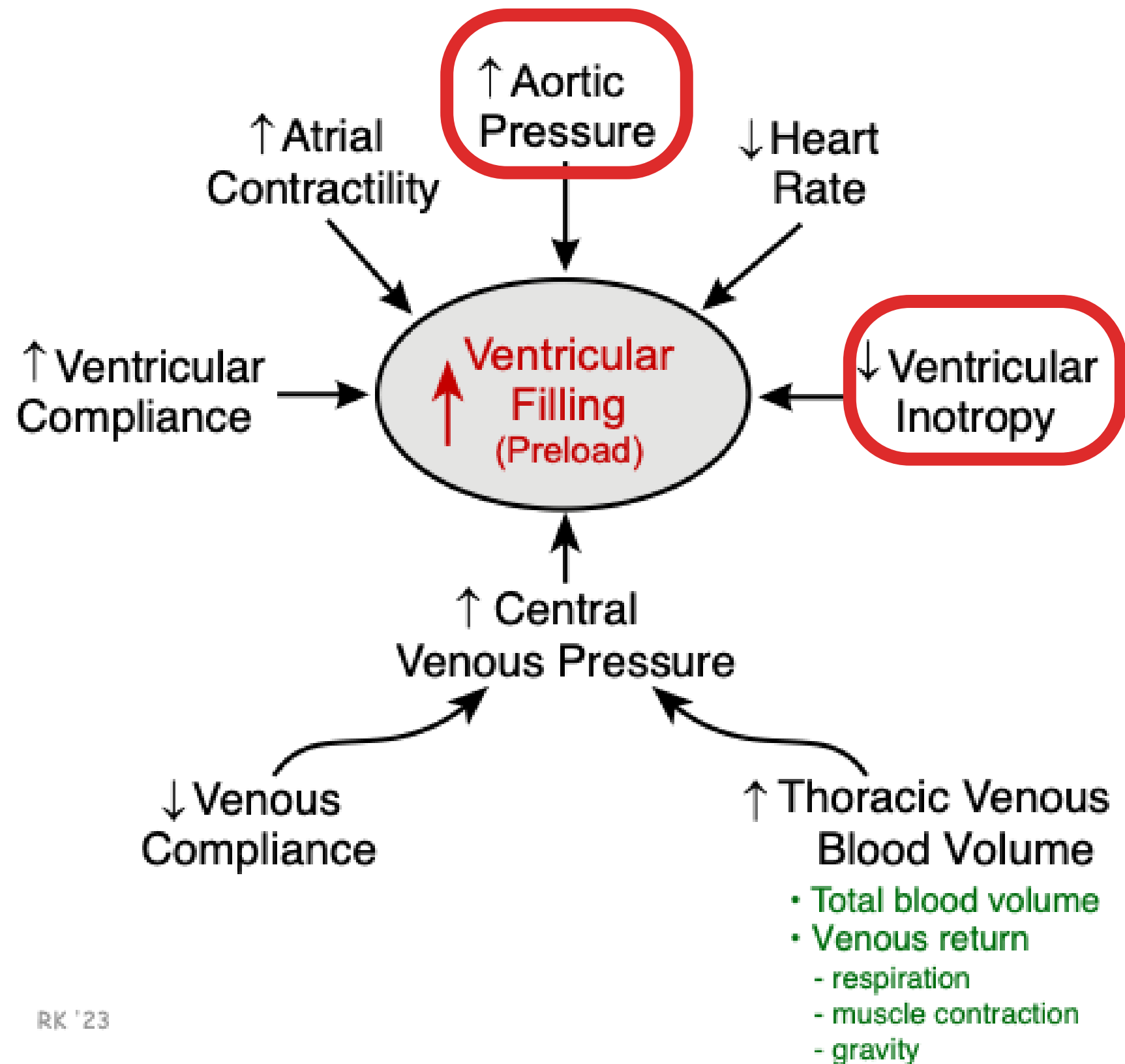


Primary Preload Regulation (direct effect)

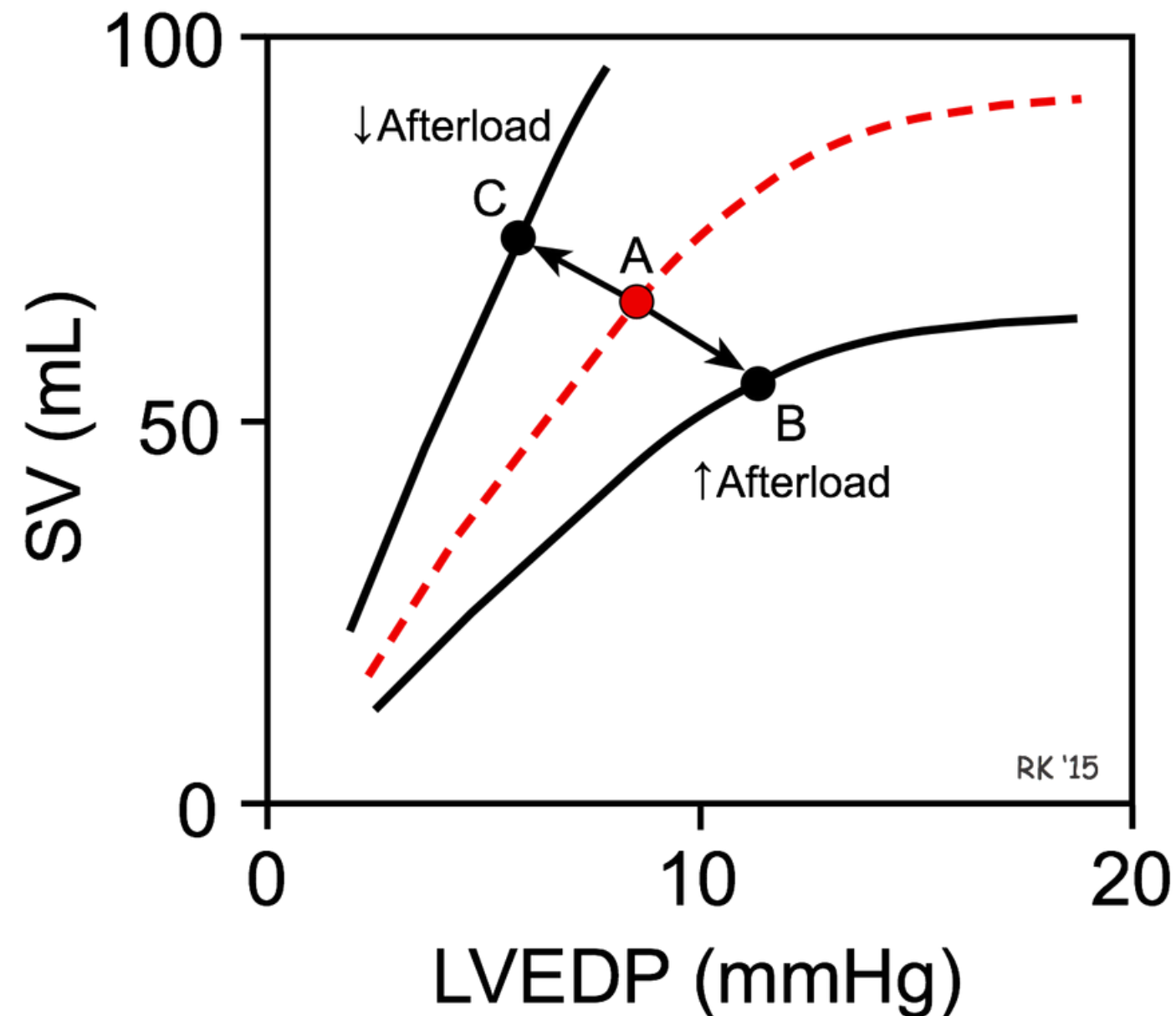
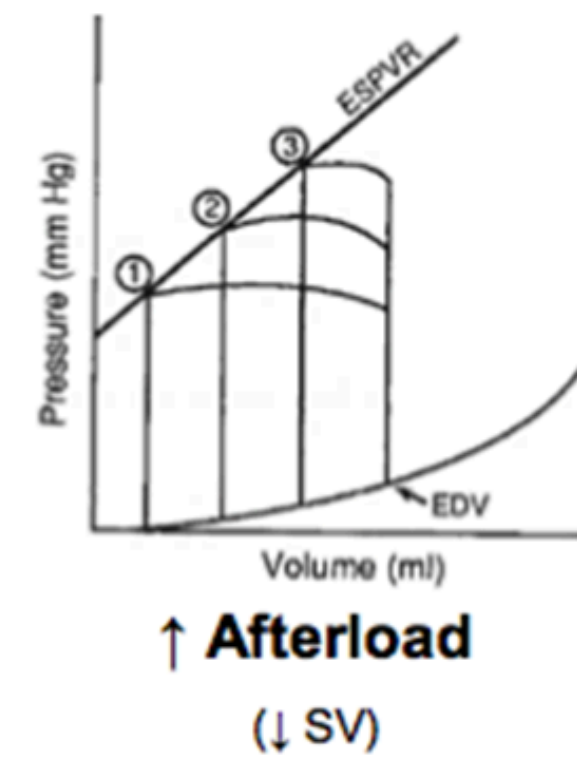


- **↑ Central venous pressure** → ↑ venous return
- **↑ Ventricular compliance** → greater filling at the same pressure
- **↑ Atrial contraction** → ↑ ventricular filling
- **↓ Heart rate** → ↑ filling time

Secondary Preload Regulation (Indirect effect)



- **↑ Aortic pressure (afterload) → ↓ SV → ↑ ESV → ↑ preload**
- **Valve defects (stenosis or insufficiency) → ↑ preload**
- **↓ Ventricular contractility (inotropy) → ↑ ESV → ↑ preload**



Effect of Afterload on Stroke Volume

↑ **Afterload** → ↓ Stroke Volume → ↑ **LVEDP**

- ↑ arterial pressure → **harder for LV to eject blood**
- ↓ fiber shortening velocity → ↓ **ejection rate**
- ↑ **ESV** → more blood remains in LV → ↑ **LVEDP** on subsequent filling

↓ **Afterload** → ↑ Stroke Volume → ↓ LVEDP

- ↓ arterial pressure → easier LV ejection
- ↑ blood ejected → ↓ **ESV** → ↓ LVEDP

4. Definition and significance of cardiac output

- **Cardiac Output (CO)** = amount of blood pumped by the heart per minute = **HR x SV**

- **What can decrease cardiac output?**

- **↓ Heart rate** or abnormal rhythm → impaired ventricular filling

- Atrial fibrillation

- **↓ Stroke volume** → e.g., ↓ contractility, ↓ preload, ↑ afterload

- **Heart disease** → myocardial infarction, aortic stenosis

- **Constrictive pericarditis** → impaired ventricular filling

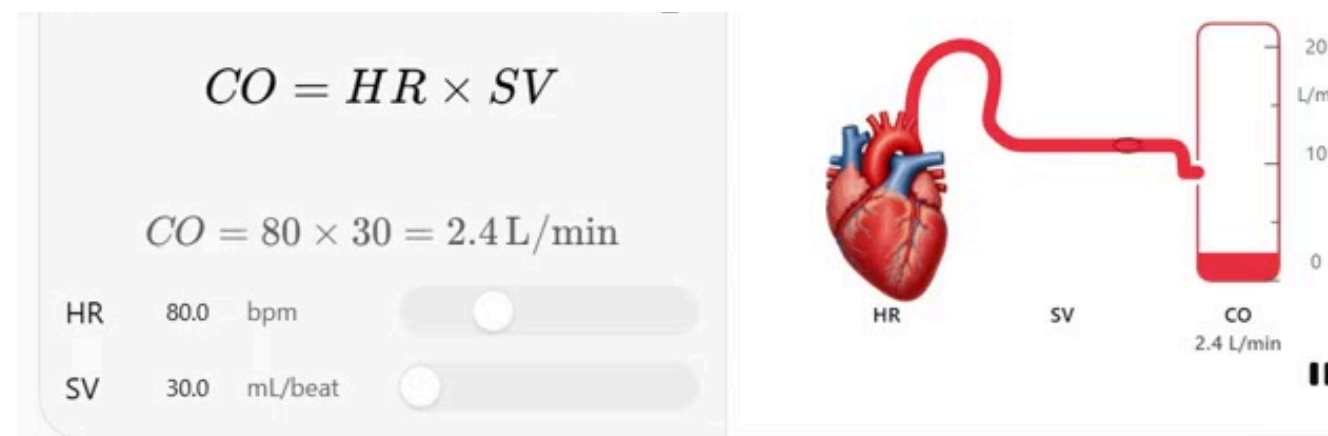
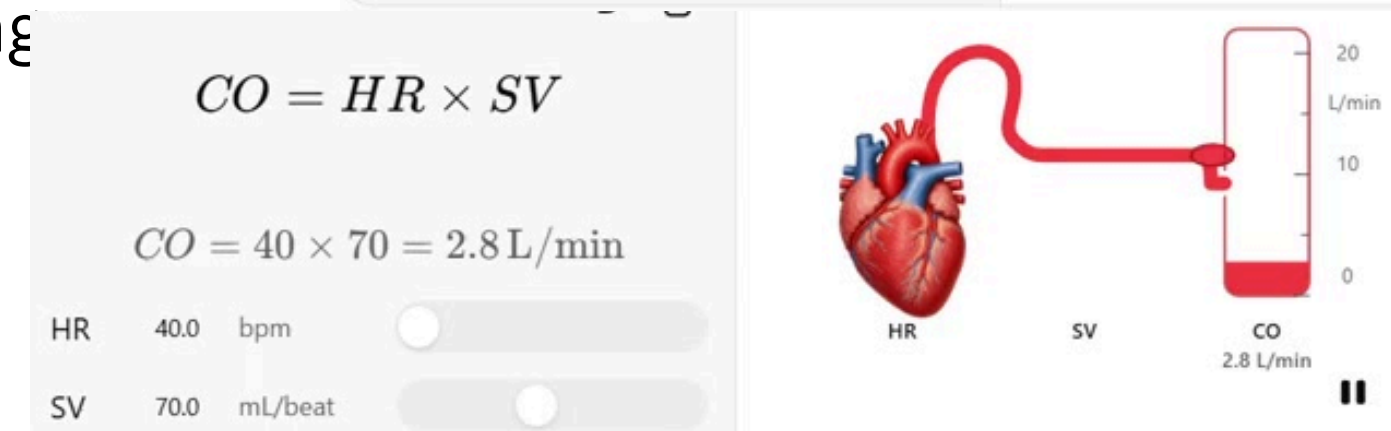
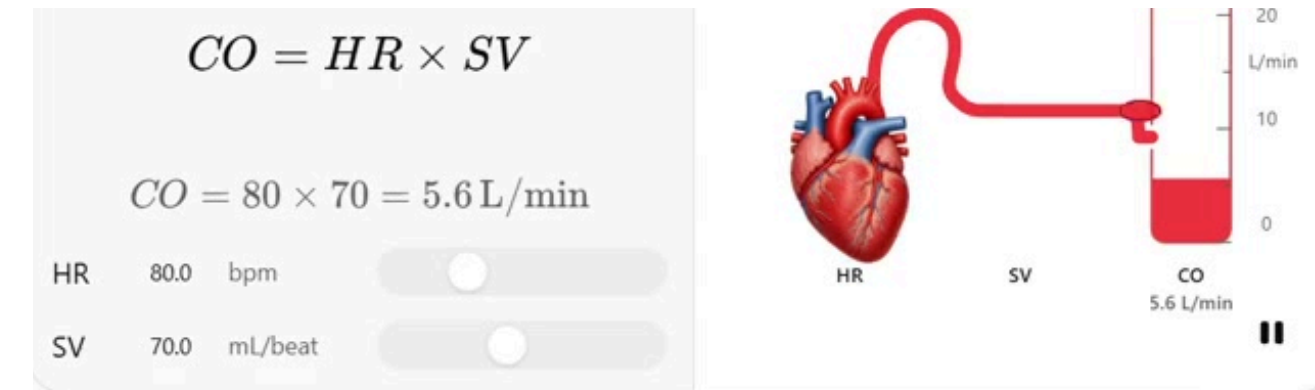
- **Symptoms of ↓ Cardiac Output**

- Fatigue / ↓ exercise capacity

- Shortness of breath

- Peripheral edema

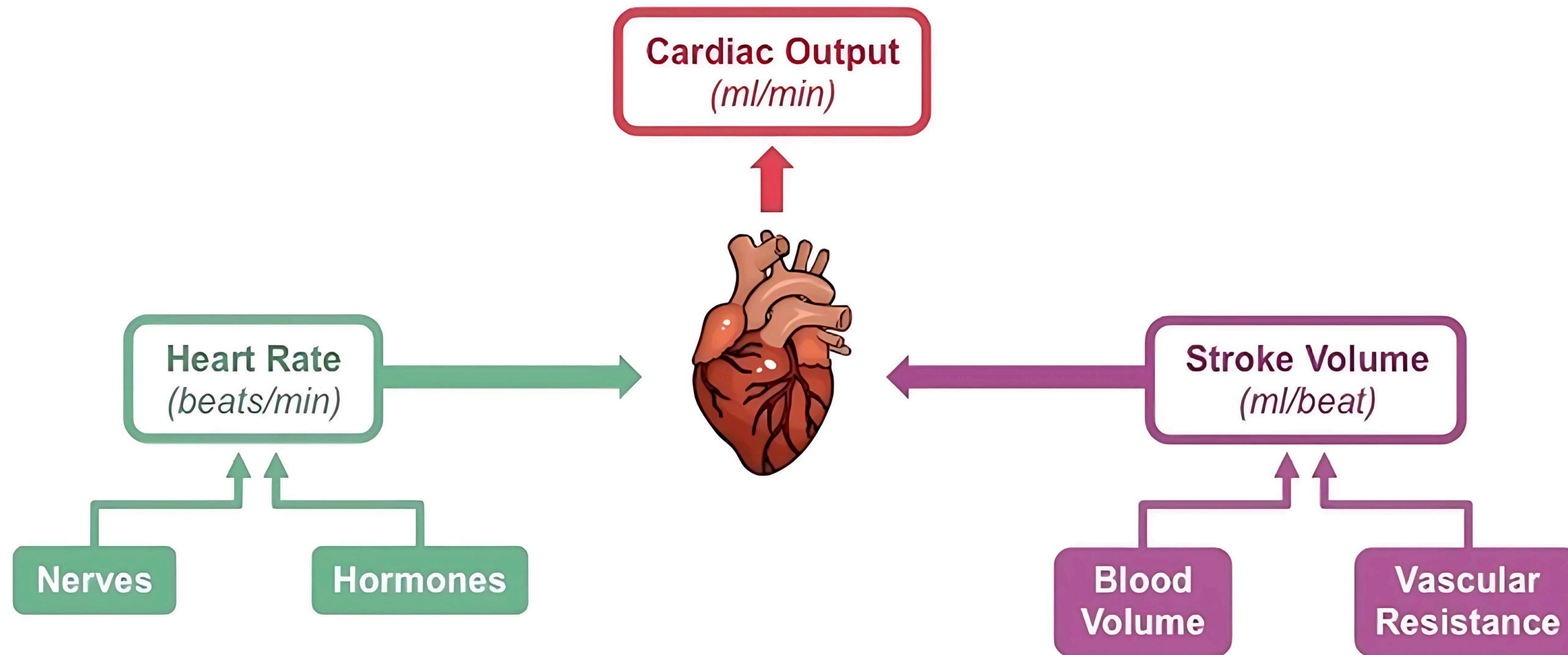
- Nausea or abdominal discomfort



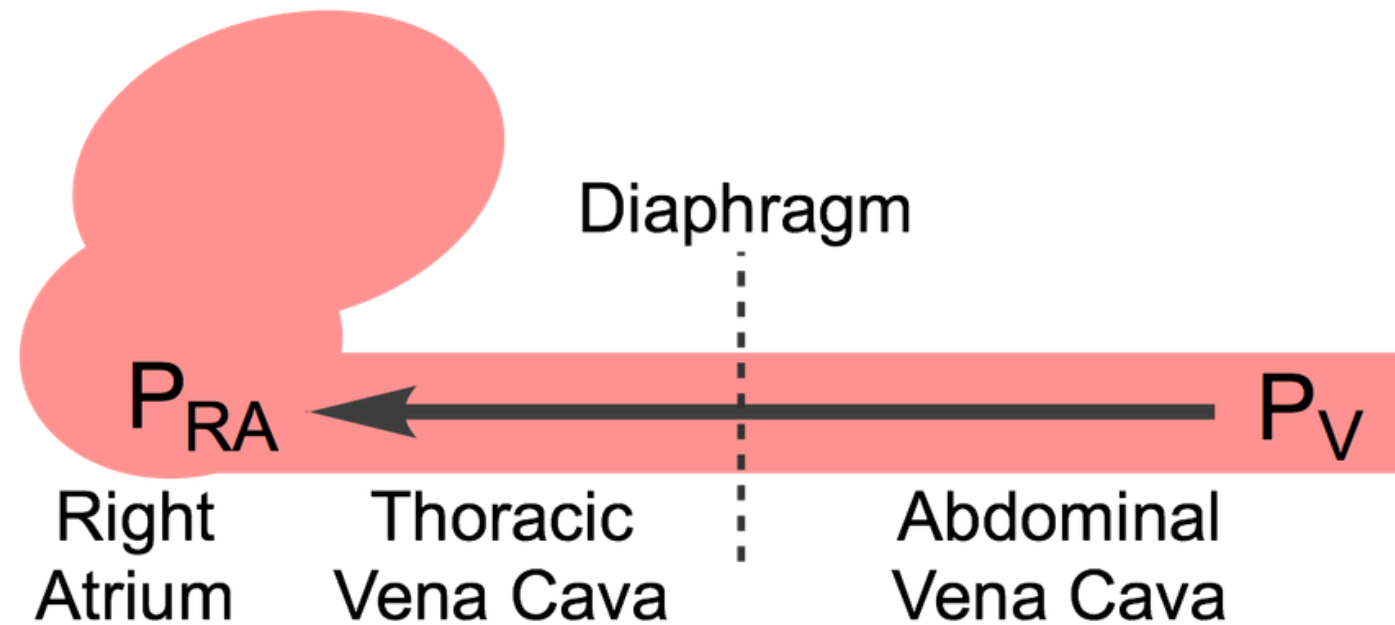
5. Regulation of cardiac output

5.1 Heart rate and factors affecting heart rate

5.2 Stroke volume and factors affecting stroke volume



6. Definition and significance of the venous return



$$VR = \frac{P_V - P_{RA}}{R_V}$$

RK '15

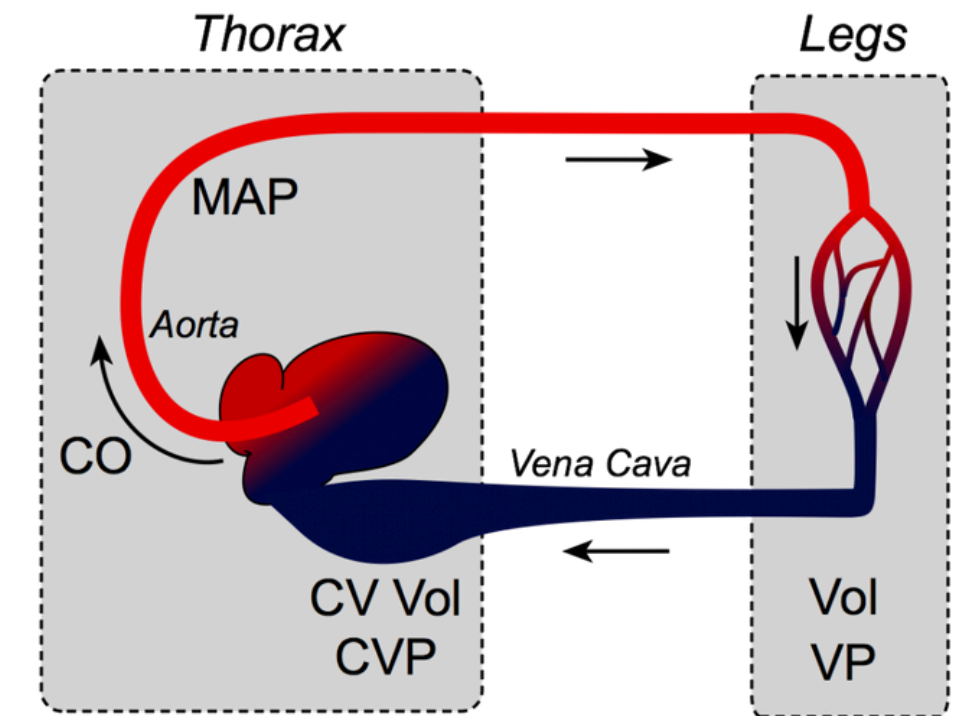
- $\uparrow$ **Venous pressure** $\rightarrow \uparrow$ pressure gradient $\rightarrow \uparrow$ VR
- $\downarrow$ **Right atrial pressure** $\rightarrow \uparrow$ pressure gradient $\rightarrow \uparrow$ VR
- $\uparrow$ Venous resistance $\rightarrow \downarrow$ VR

VR is driven by the **pressure gradient** from the systemic veins to the right atrium.

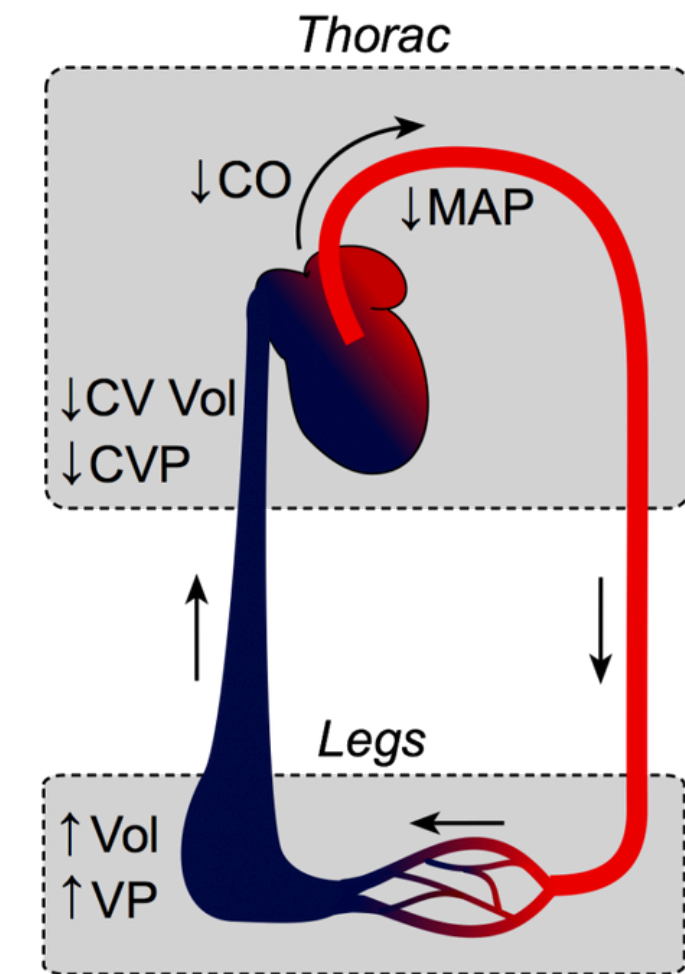
7. Factors that influence venous return

7.1 Postural change: Orthostatic Hypotension

- Standing up → gravity shifts ~500–1000 mL blood to the lower body
 - ↓ Venous return → ↓ preload
 - ↓ Stroke volume → ↓ cardiac output
 - → ↓ Blood pressure
- Normally, the **baroreceptor reflex** increases HR and vasoconstriction to maintain BP.
 - ↓ SBP $\geq$ 20 mmHg OR ↓ DBP $\geq$ 10 mmHg
 - Occurs within 3 minutes of standing



Supine



Standing

7. Factors that influence venous return

7.2 Skeletal muscle pump

- skeletal muscles help maintain venous return and consequently cardiac output by **compressing underlying veins** in order to increase blood flow back to the heart



1. **At rest:** both proximal & distal valves are open, blood is flowing due to 16mm Hg blood pressure of veins
2. A muscle **Contraction** pushes blood through top valve '**milking**' it towards heart. Because bottom valve blocks blood flow downwards.
3. upon **Relaxation** recently milked blood cannot flow back down because top valve closes to block it. **Middle section is 'empty'**, thus BP is higher in foot. **Blood from foot opens bottom valve and fills vein.**

7. Factors that influence venous return

7.3 Venacava compression

- An increase in the resistance of the vena cava, as occurs when the thoracic vena cava becomes compressed during a Valsalva maneuver or during late pregnancy, decreases return.

7.4 Respiration activity

- Respiration strongly affects venous return, especially from **thoracic veins**.
 - Inspiration → ↓ intrathoracic pressure → ↑ blood flow into the right atrium → ↑ venous return
 - Expiration → intrathoracic pressure rises → ↓ venous return

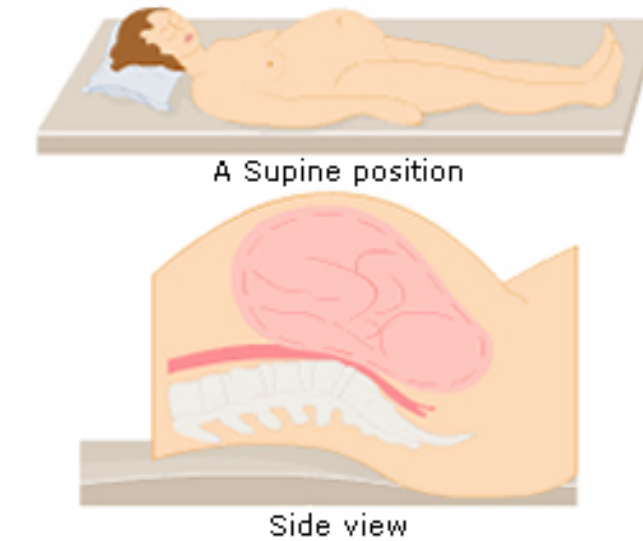


Fig 1 Supine position

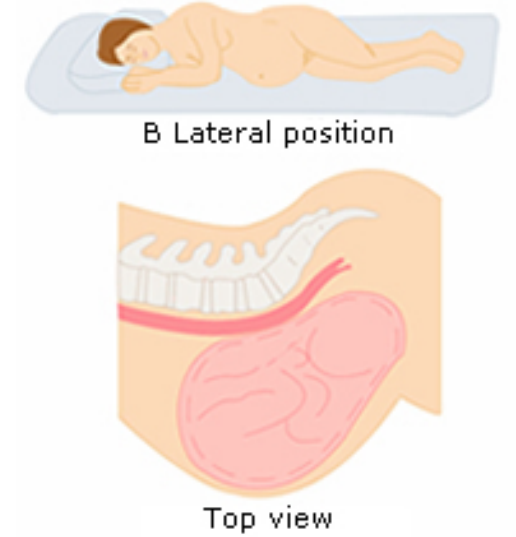
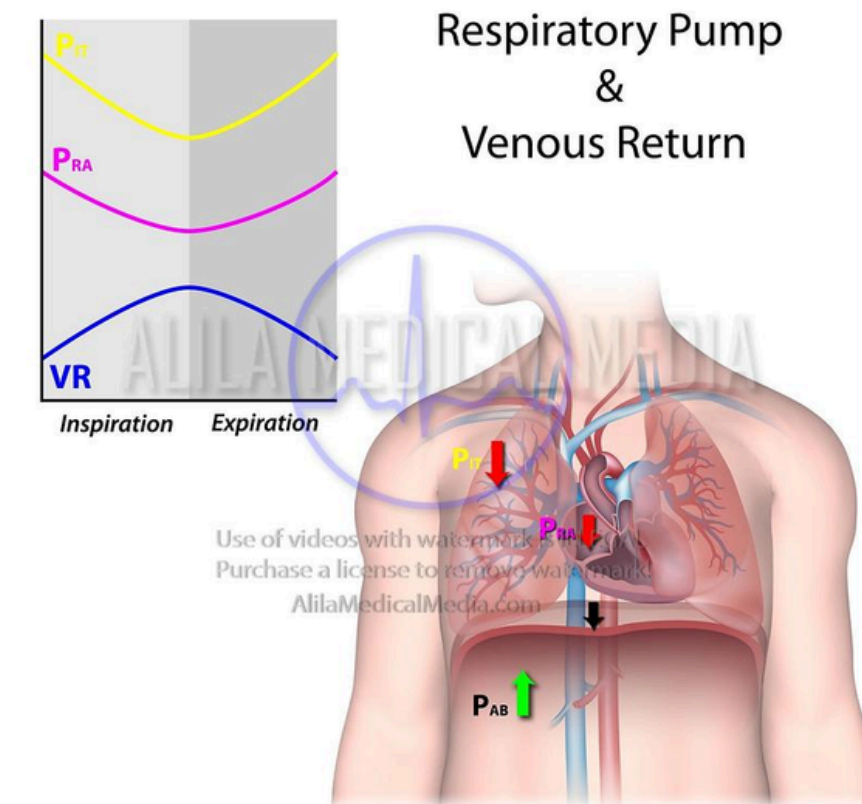


Fig 2 Lateral position



8. Alterations in cardiac output and venous return due to the pathology of the cardiovascular system and extracellular fluid electrolyte abnormalities

● Hypercalcemia ($\uparrow \text{Ca}^{2+}$)

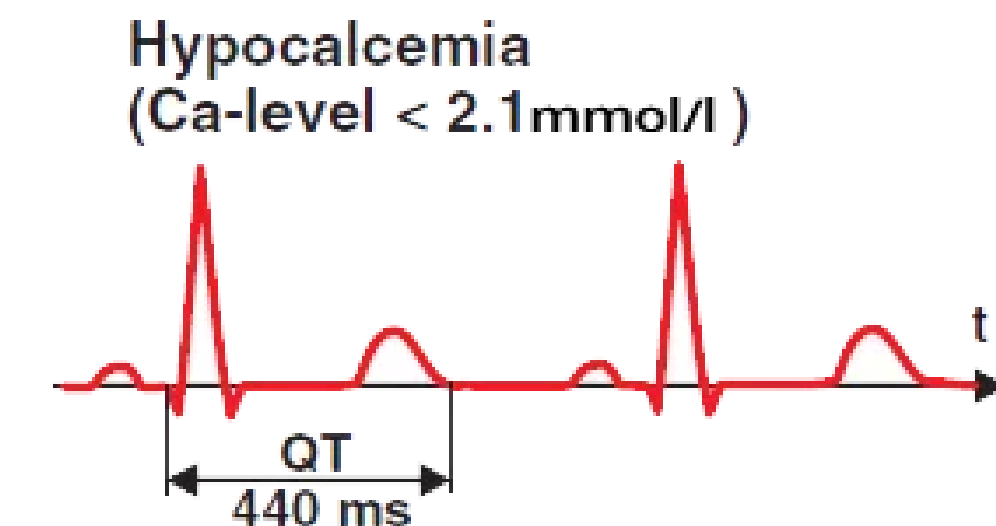
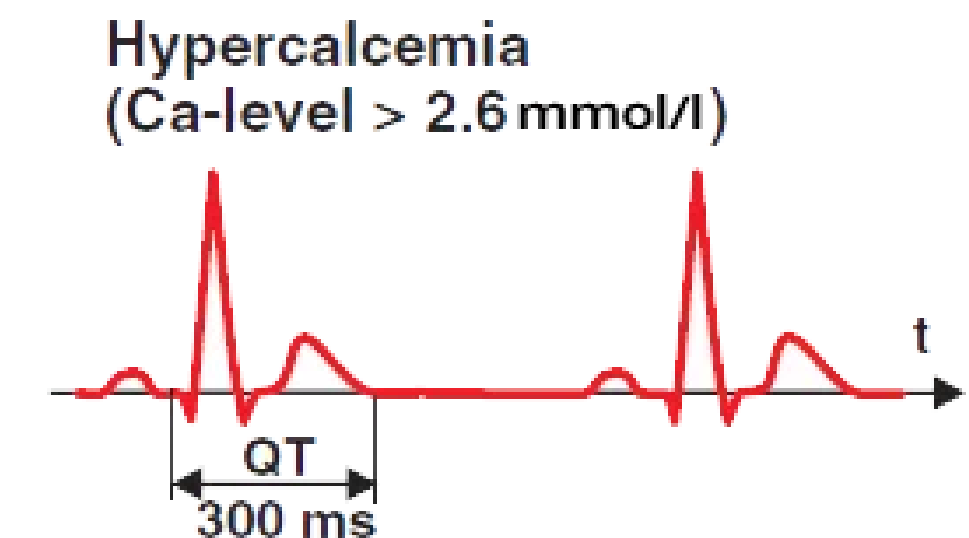
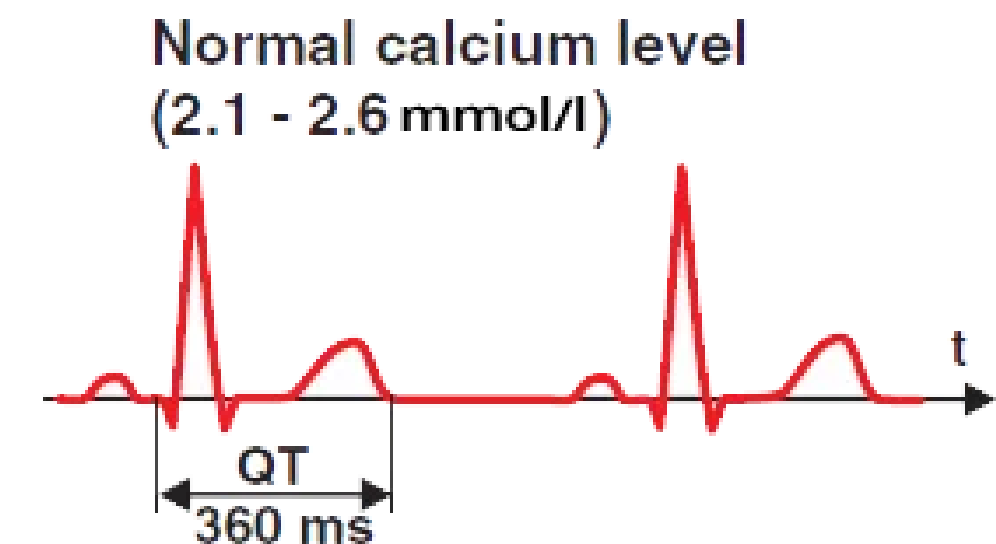
- $\uparrow$ extracellular Ca^{2+} → **altered L-type Ca^{2+} channel kinetics**
- $\downarrow$ duration of phase 2 (plateau)
 - **earlier ventricular repolarization** → **Shortened QT interval**

● Hypocalcemia ($\downarrow \text{Ca}^{2+}$)

- $\downarrow$ extracellular Ca^{2+} → prolonged plateau phase
 - → delayed ventricular repolarization → Prolonged QT interval

Electrolyte imbalance → altered cardiac action potential → altered

ECG → impaired cardiac function → possible changes in cardiac output.



8. Alterations in cardiac output and venous return due to the pathology of the cardiovascular system and extracellular fluid electrolyte abnormalities

● Hyperkalemia ($\uparrow K^+$)

- $\uparrow$ extracellular K^+ $\rightarrow$ **less negative RMP**
 - Moderate $\uparrow K^+$ $\rightarrow$ $\uparrow K^+$ conductance $\rightarrow$ shorter action potential $\rightarrow$ **Peaked T waves**
 - Severe hyperkalemia $\rightarrow$ Na^+ channels become less available $\rightarrow$ $\downarrow$ conduction velocity $\rightarrow$ **P wave flattening/disappearance + QRS widening**

● Hypokalemia ($\downarrow K^+$)

- $\downarrow$ extracellular K^+ $\rightarrow$ $\downarrow$ repolarization reserve $\rightarrow$ **delayed ventricular repolarization** $\rightarrow$ prolonged action potential
- ECG: flattened T wave, prominent U wave, and prolonged QT interval

